

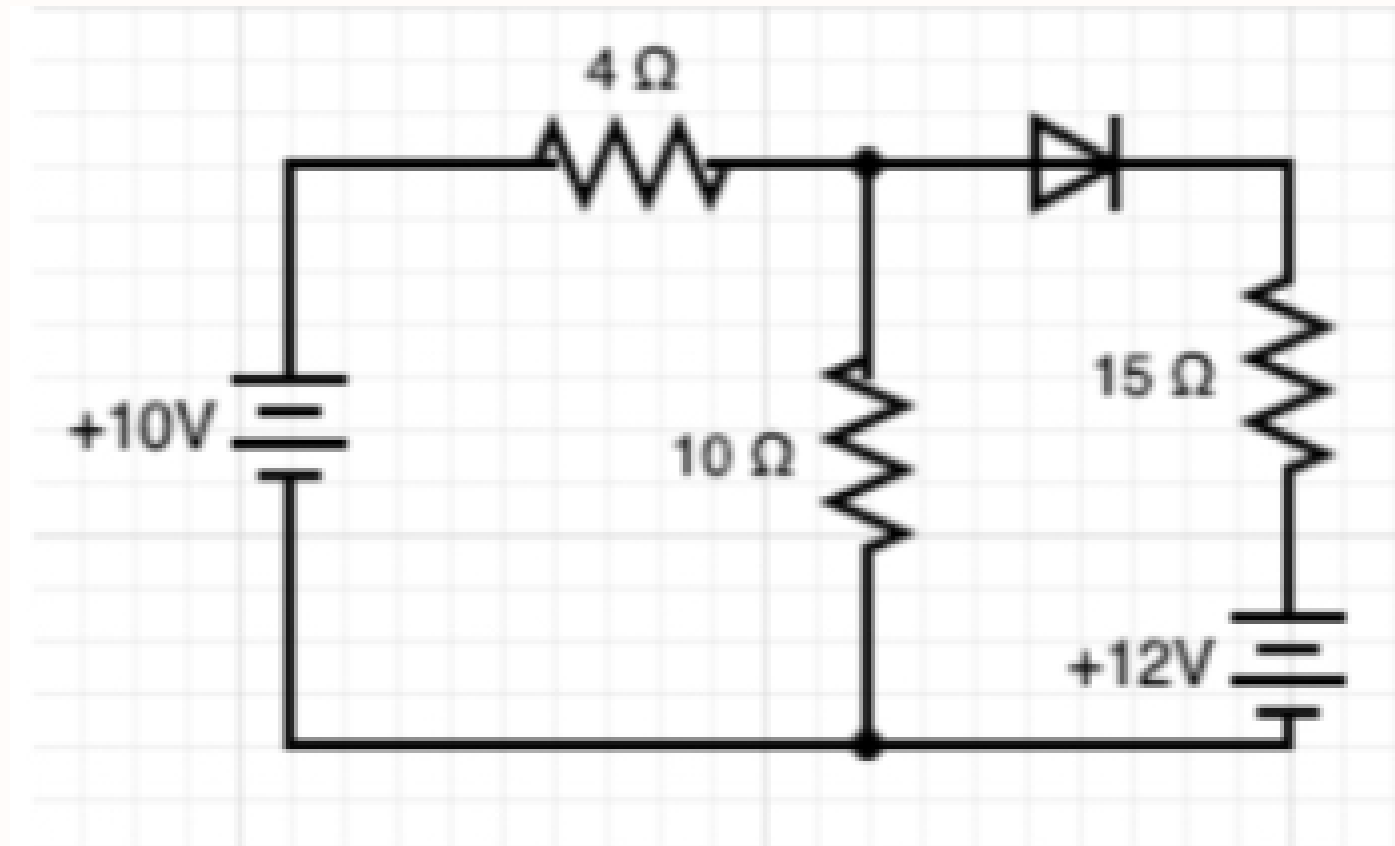


EGCI 232 Engineering Electronics

- Diode Circuits and Applications



Ideal diode circuit





Applications with diodes

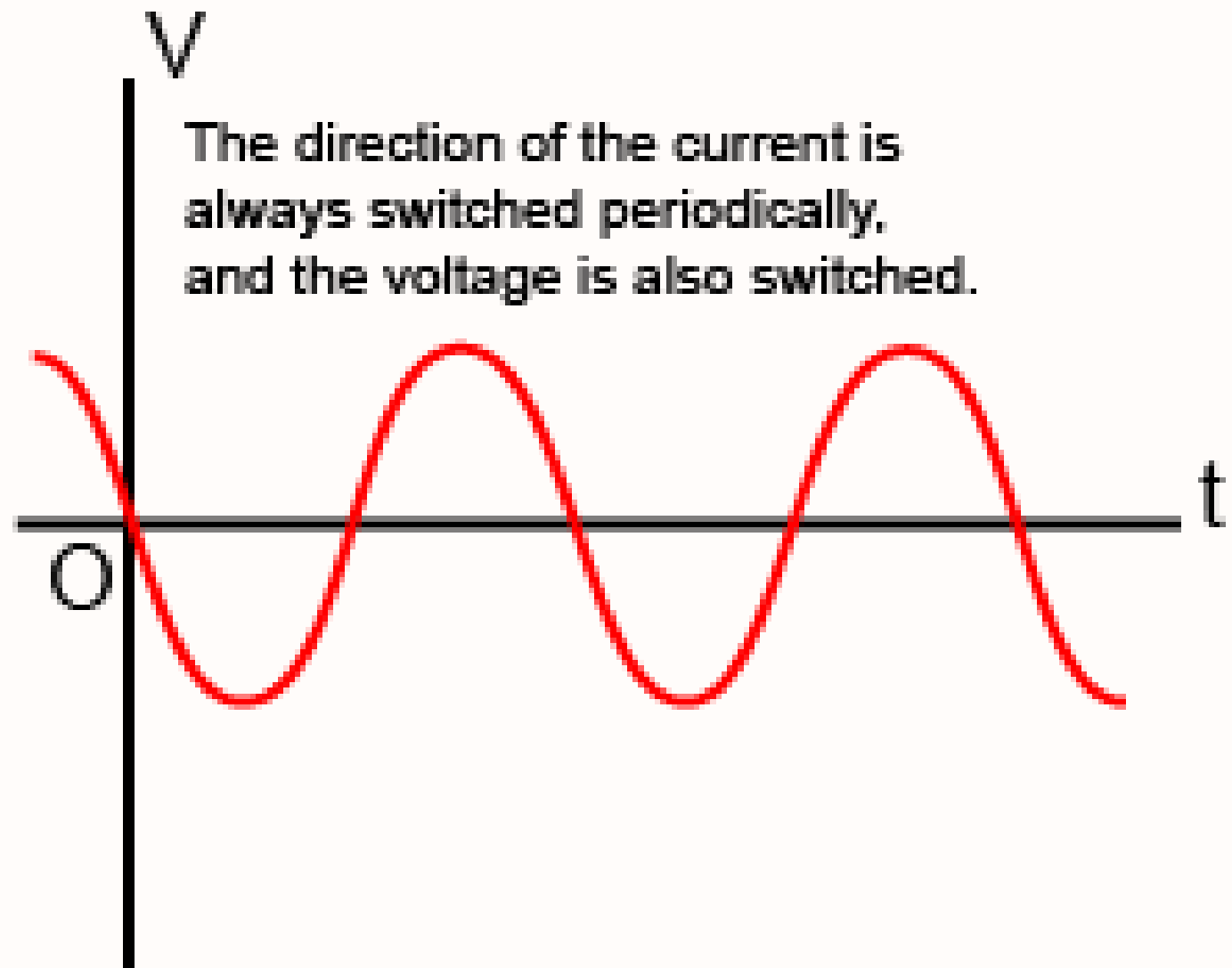
- **Rectifier circuit**
- **Clipping circuit or limiter**
- **Clamplng circuit**
- **Voltage multiplier circuit**
- **Logic gates**



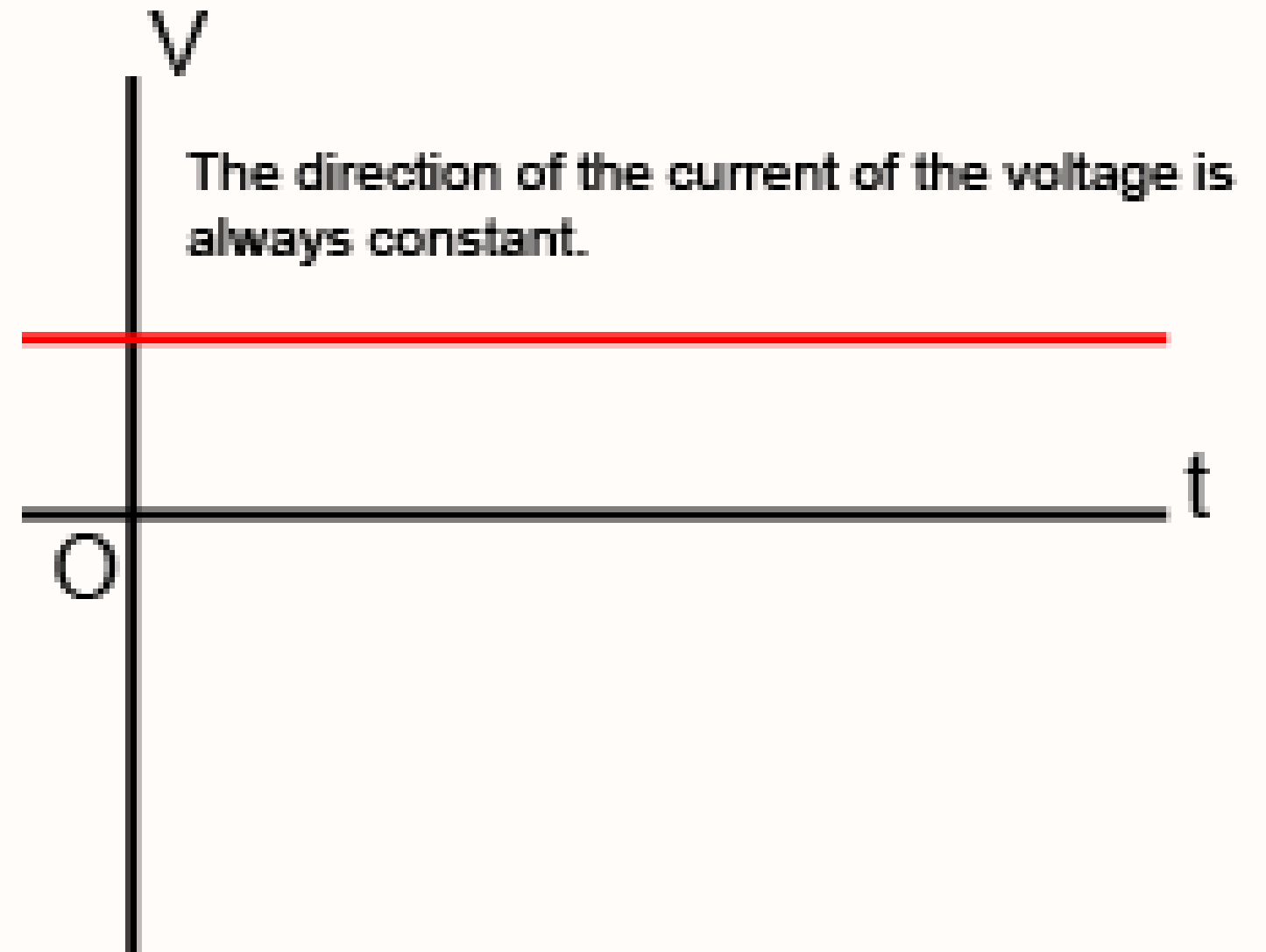
Rectifier Circuit

To convert AC voltage to DC voltage

Alternating Current (AC)



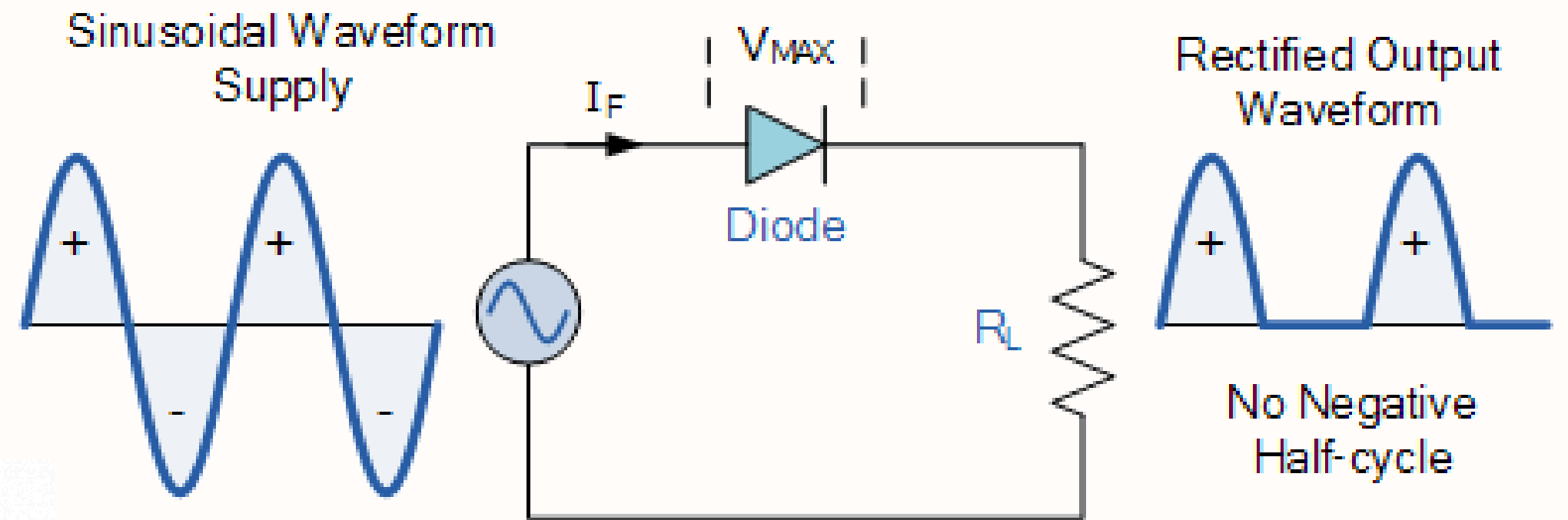
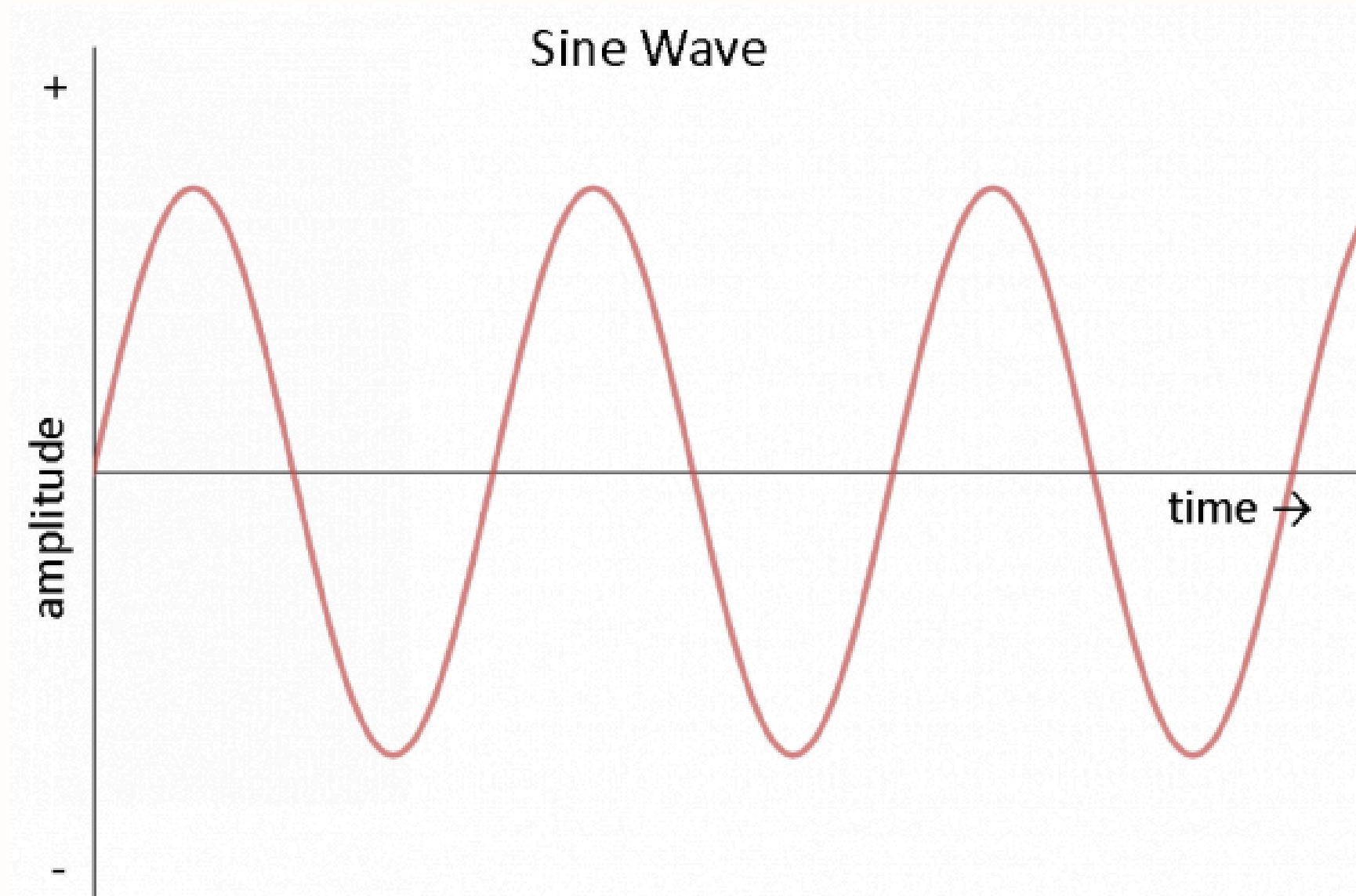
Direct Current (DC)





Rectifier Circuit

Half-wave rectifier



For half wave rectifier

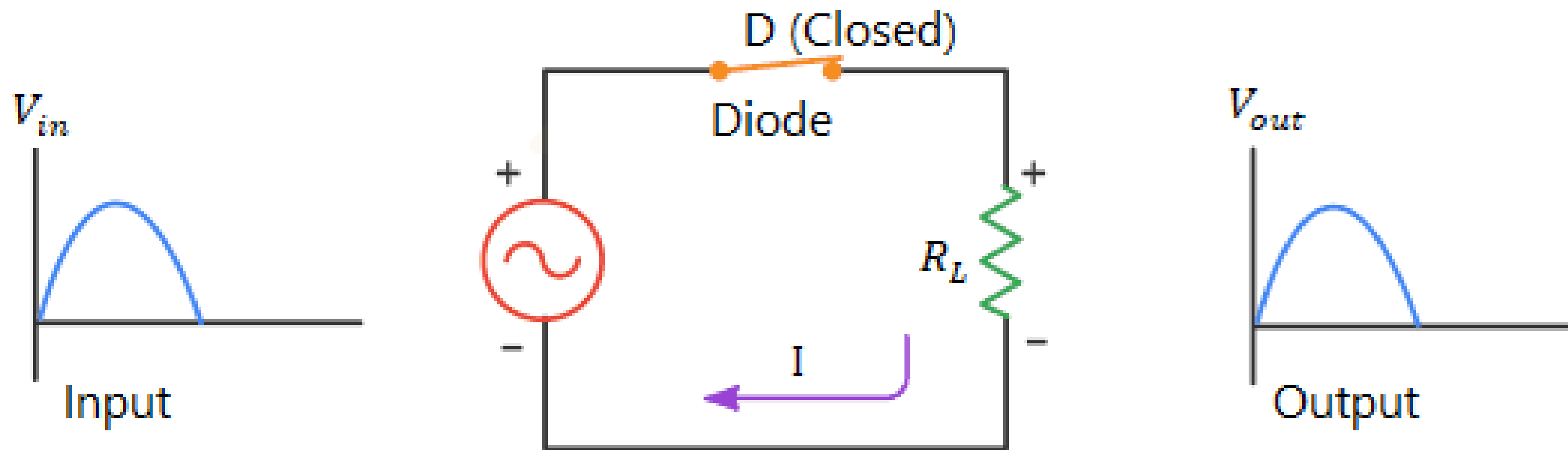
$$V_{DC} = \frac{1}{2\pi} \int_0^{\pi} V \sin \omega t \, dt$$

$$= \frac{1}{2\pi} \times 2V_M = \frac{V_m}{\pi}$$



Rectifier Circuit

Half-wave rectifier



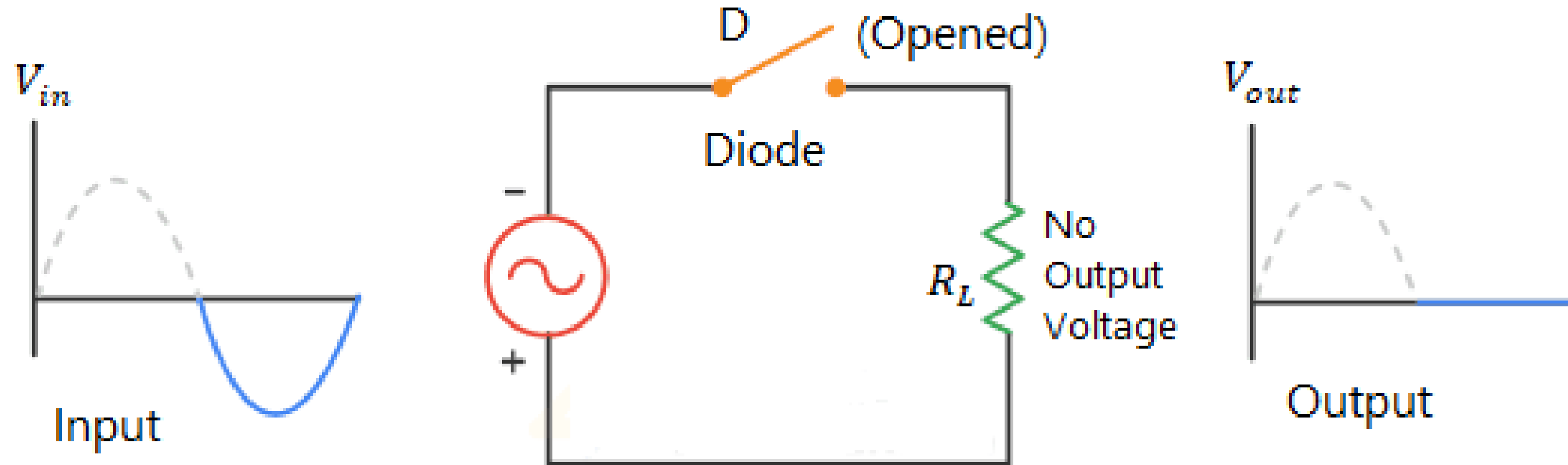
Half Wave Rectifier circuit during positive half cycle

How2Electronics



Rectifier Circuit

Half-wave rectifier



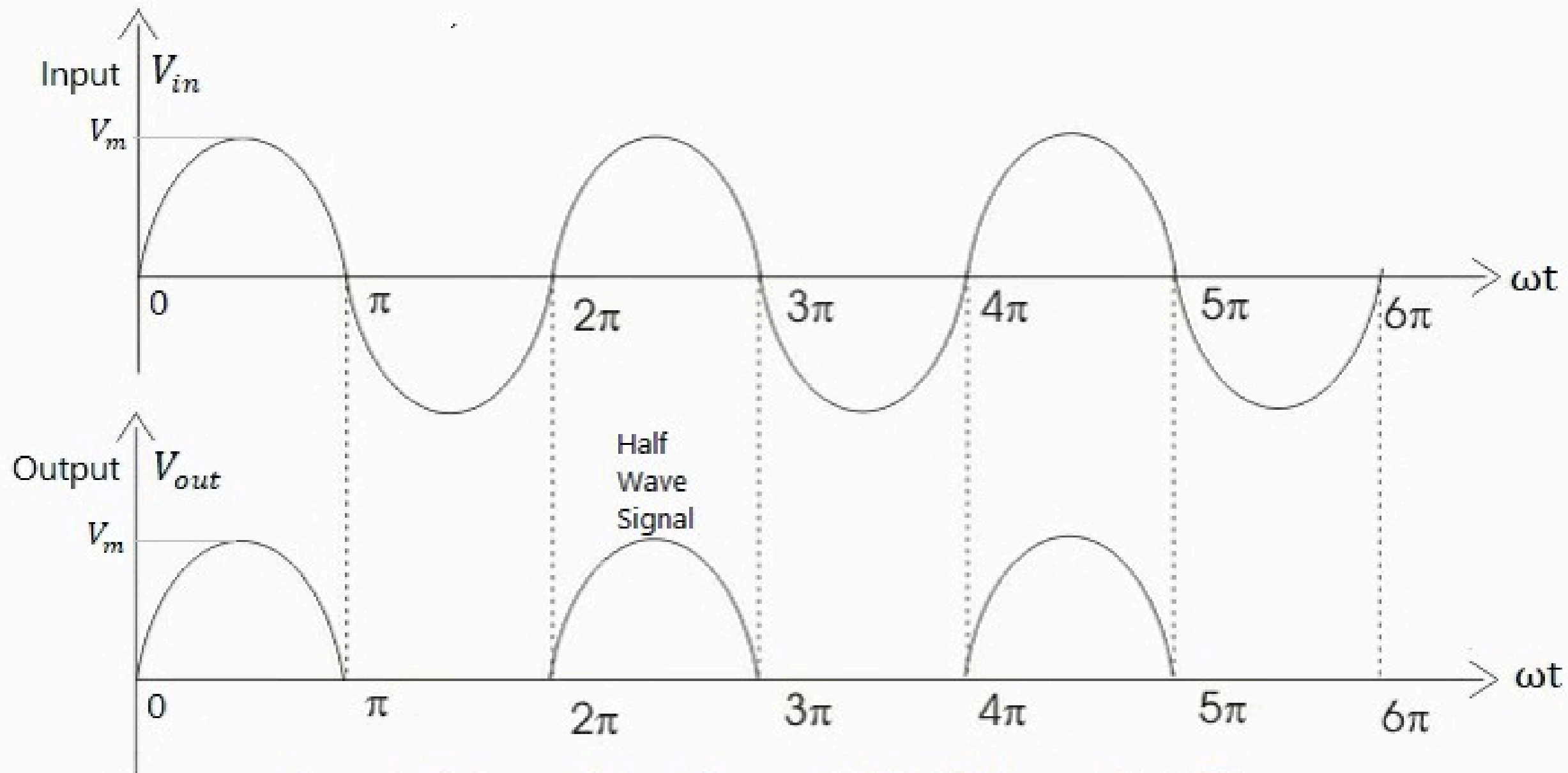
Half Wave Rectifier circuit during negative half cycle

How2Electronics



Rectifier Circuit

Half-wave rectifier

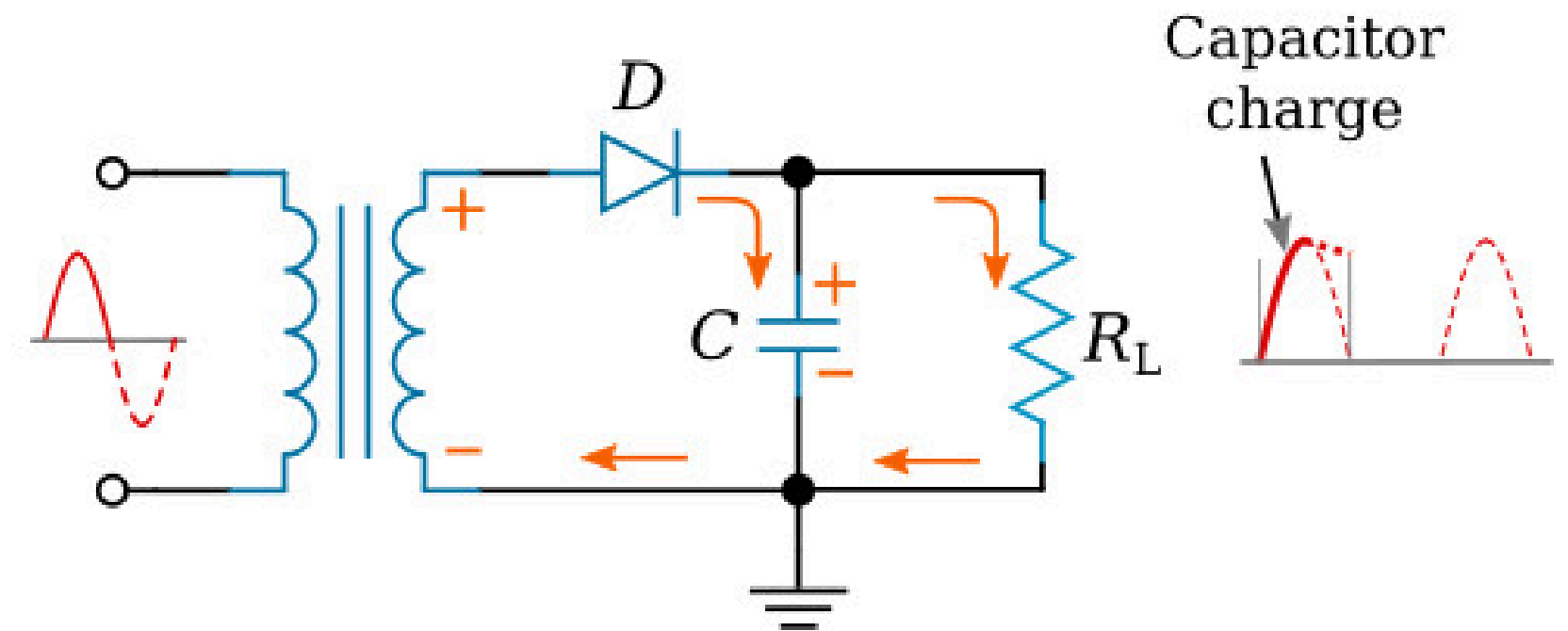
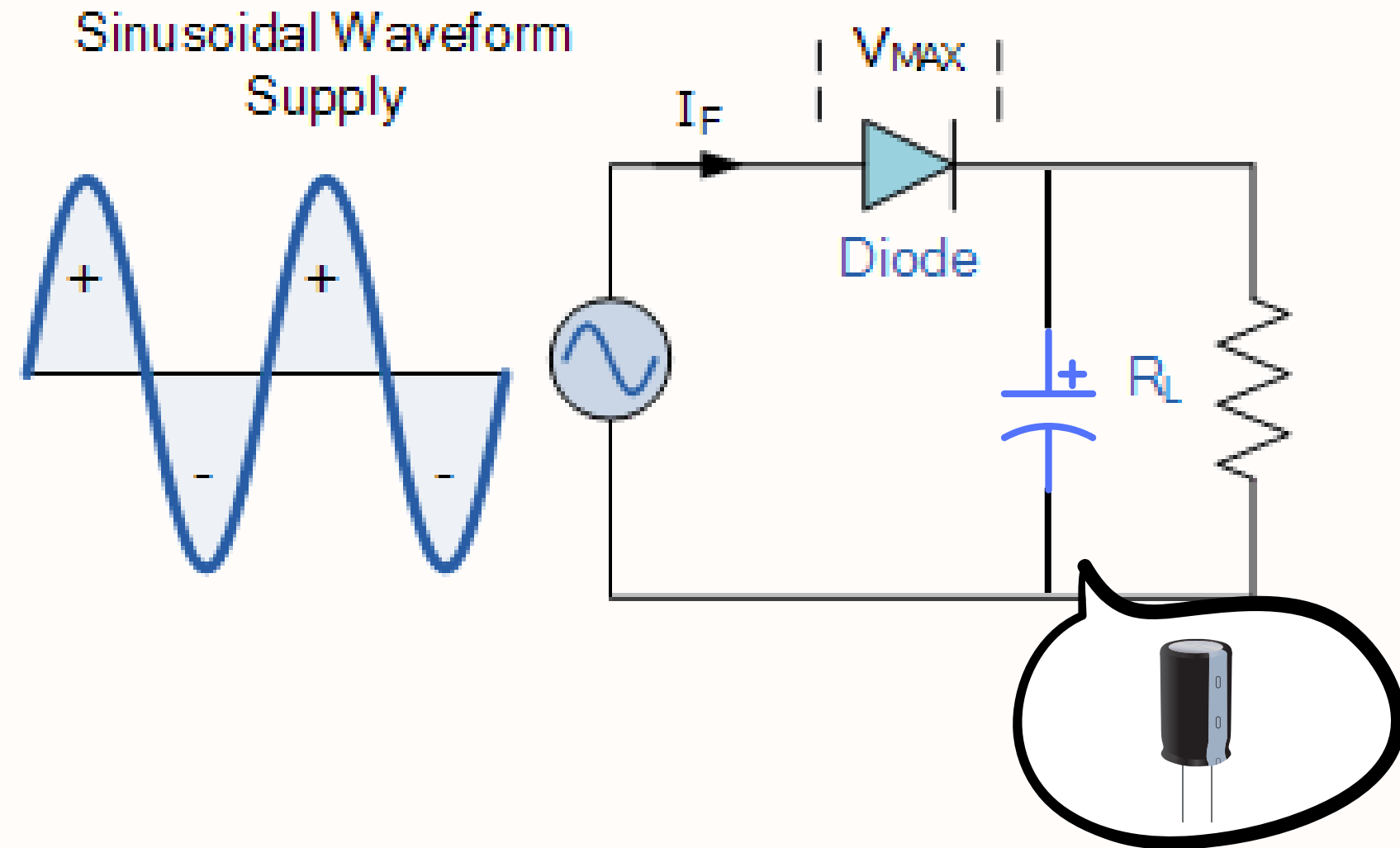


Input Output Waveform of Half Wave Rectifier

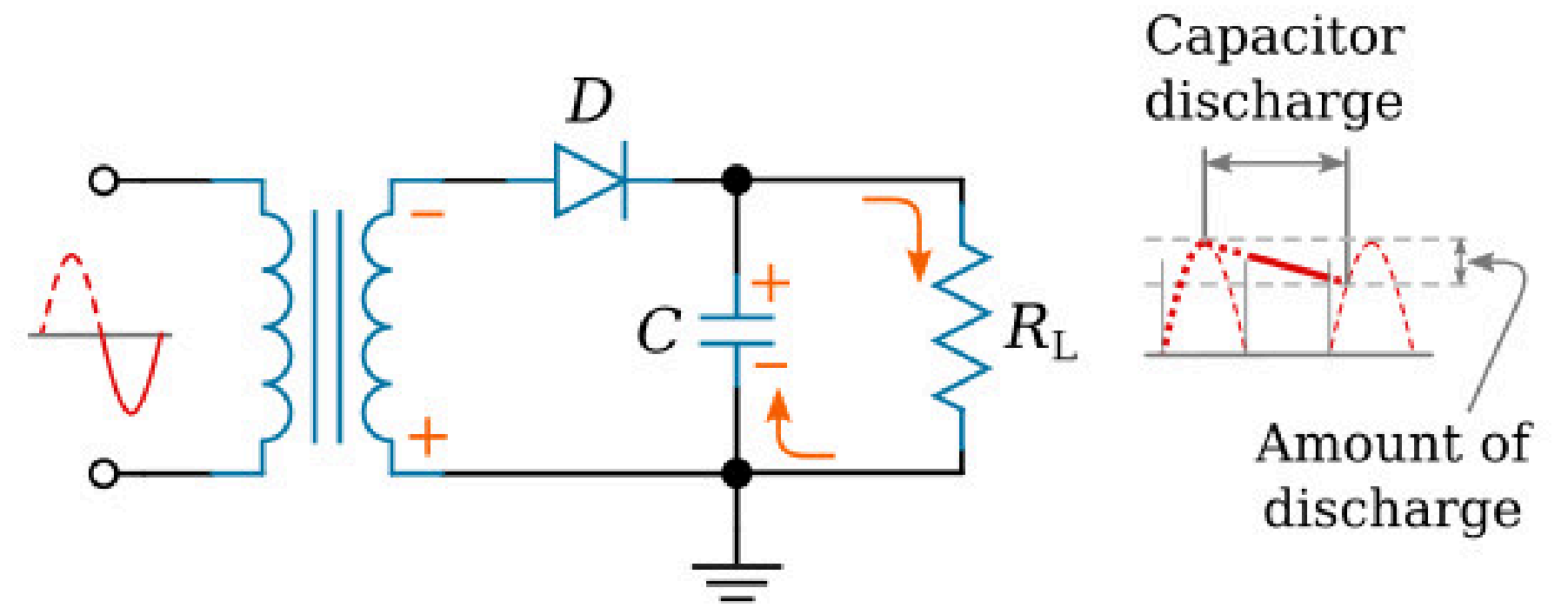


Rectifier Circuit

Half-wave rectifier (with filter)



(A) Positive half-cycle

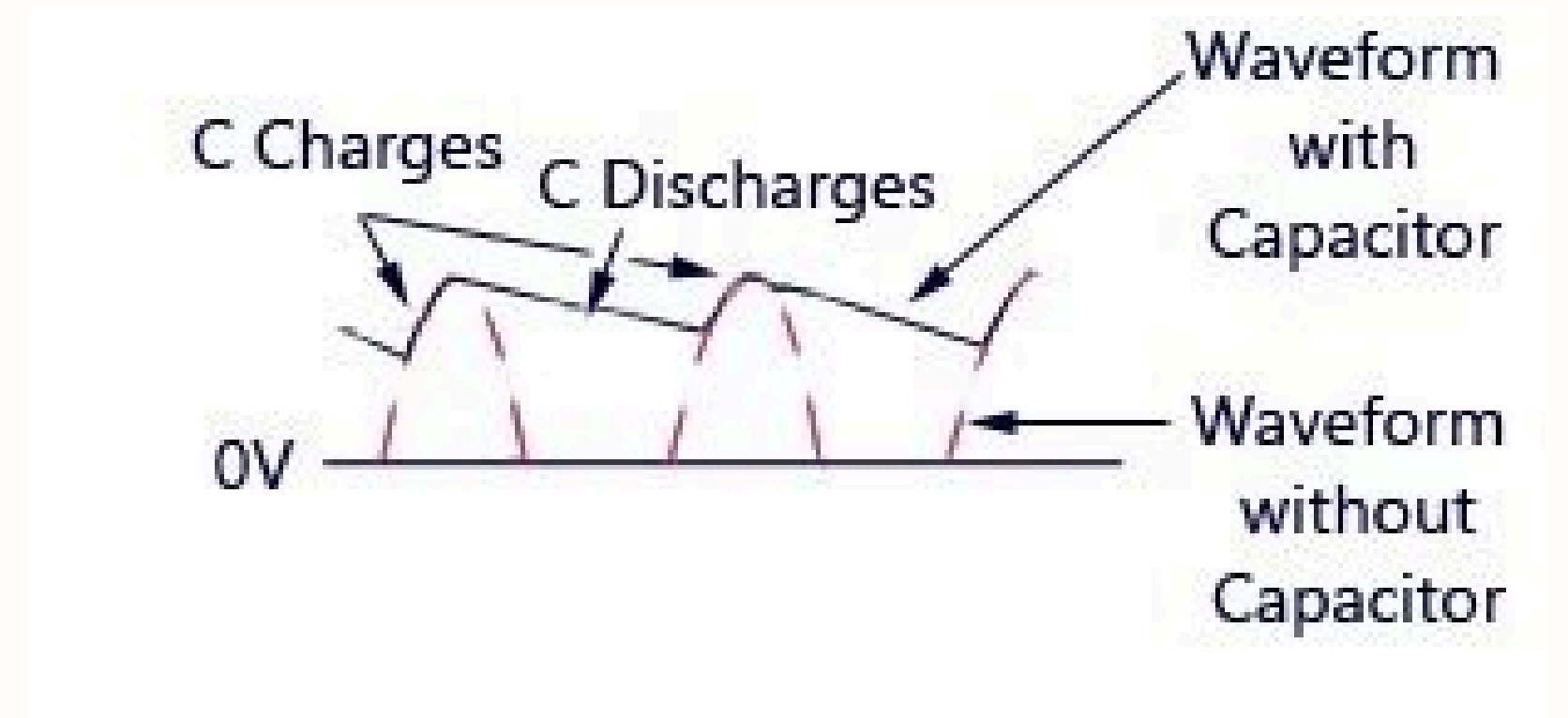
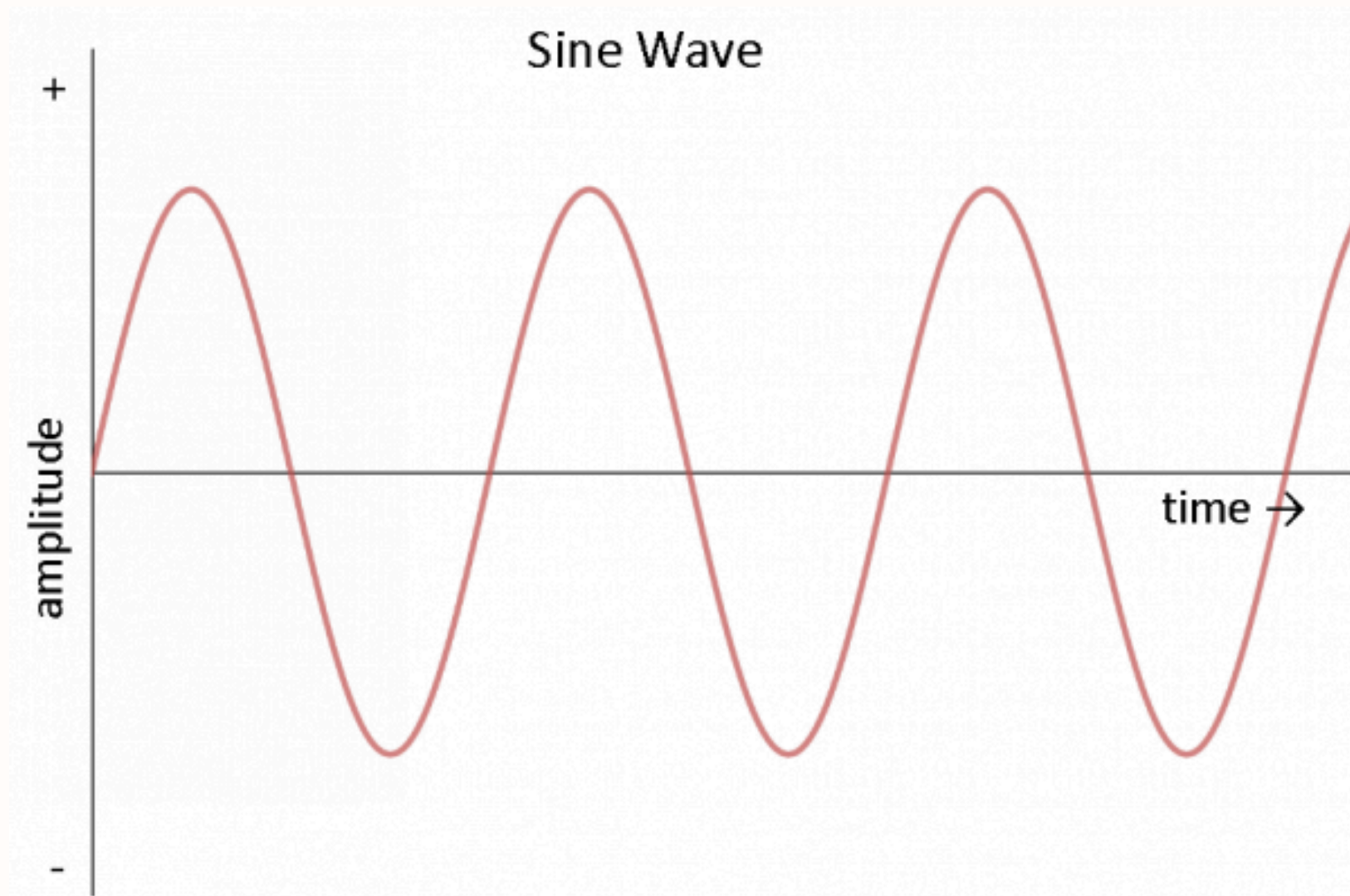


(B) Negative half-cycle



Rectifier Circuit

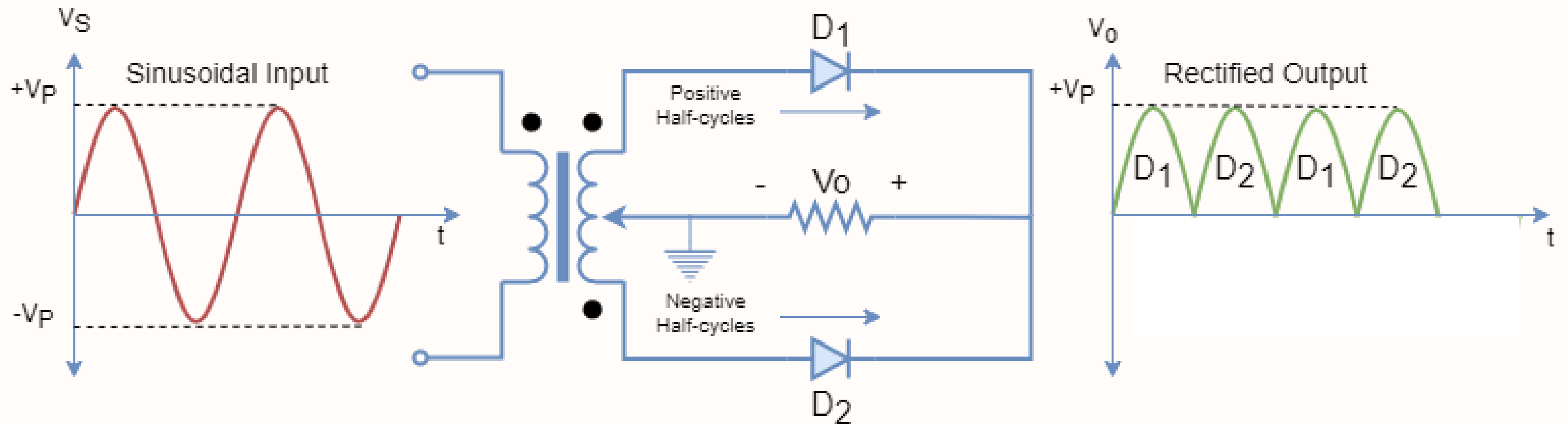
Half-wave rectifier (with filter)





Rectifier Circuit

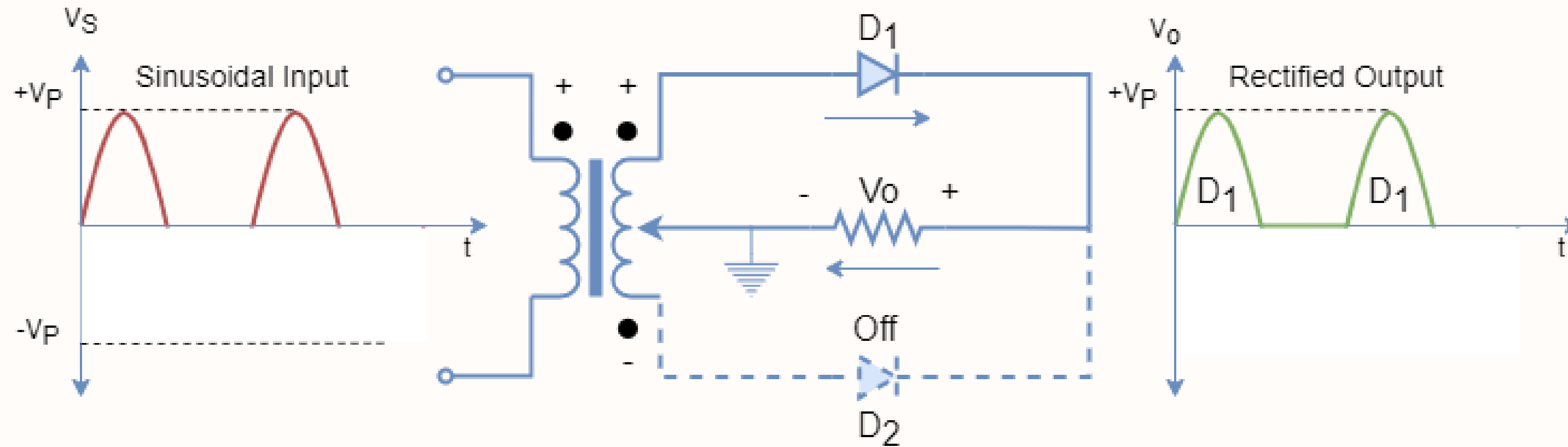
Full-wave rectifier (center-tapped)





Rectifier Circuit

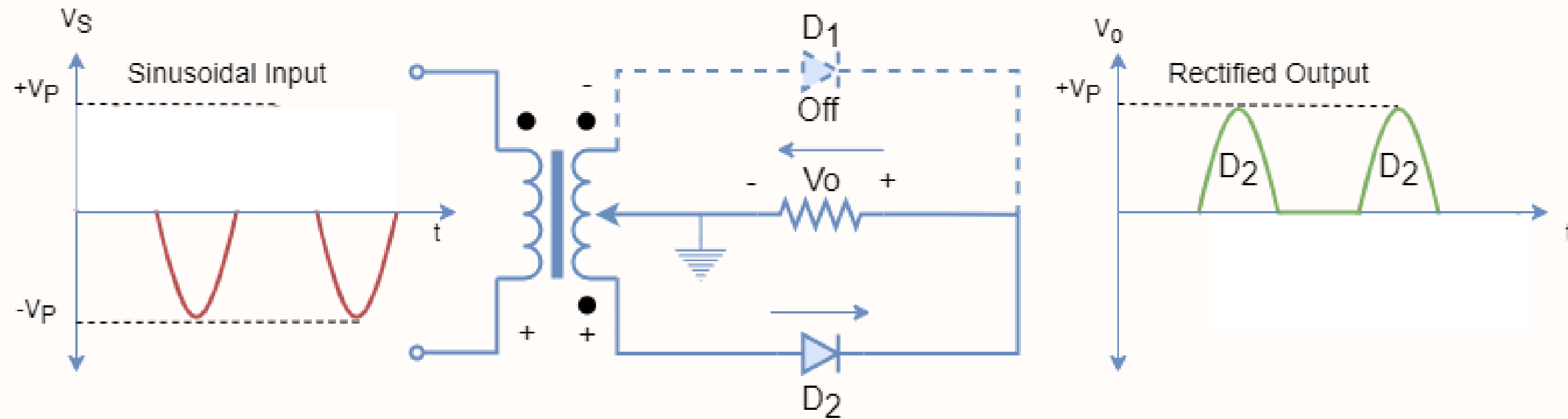
Full-wave rectifier (center-tapped)





Rectifier Circuit

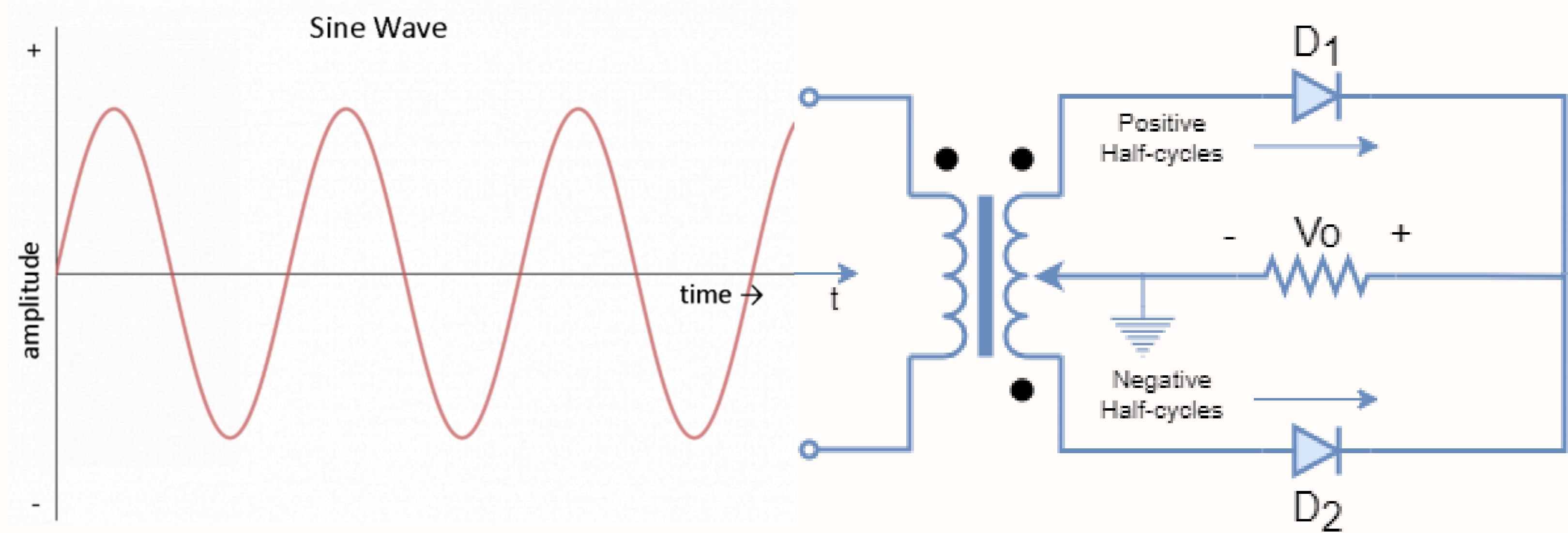
Full-wave rectifier (center-tapped)





Rectifier Circuit

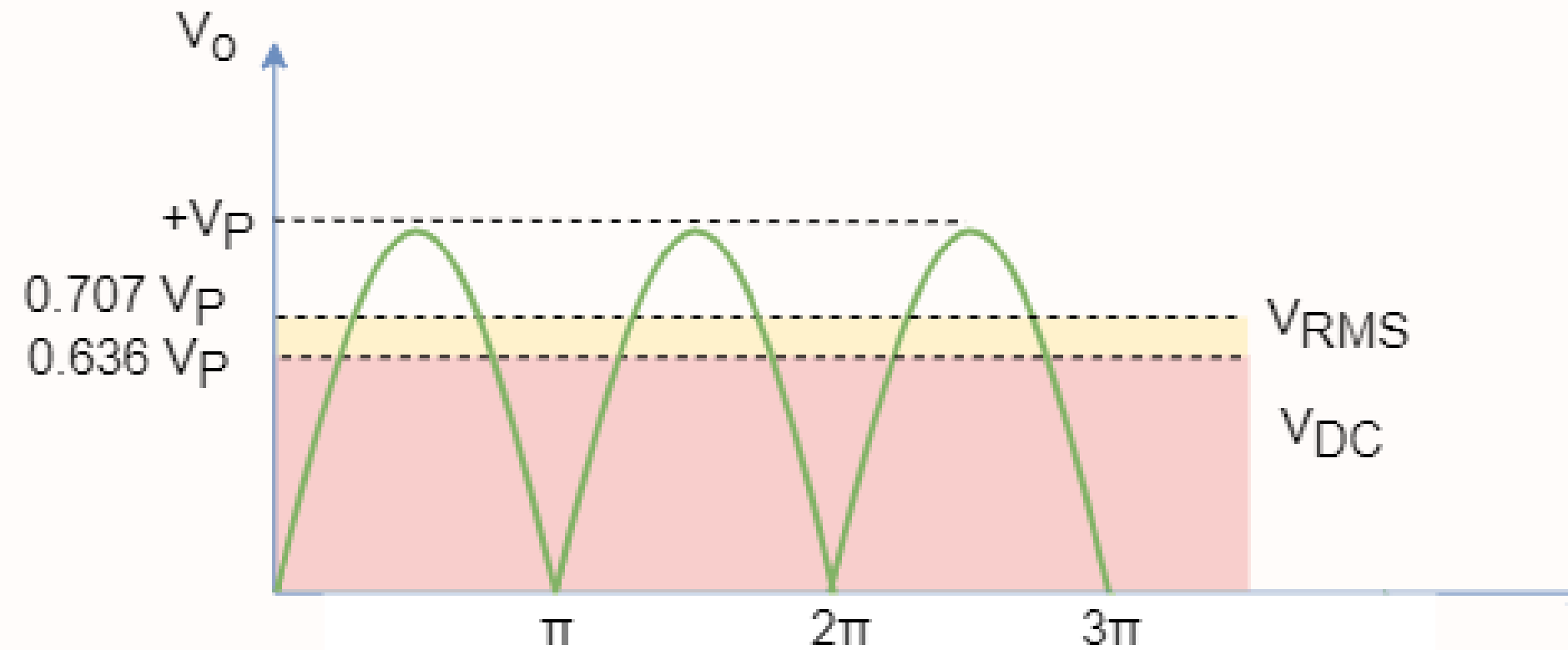
Full-wave rectifier (center-tapped)





Rectifier Circuit

Full-wave rectifier (center-tapped)



$$V_{AVG} = \frac{1}{T} \int_0^T V_P \sin \theta d\theta$$

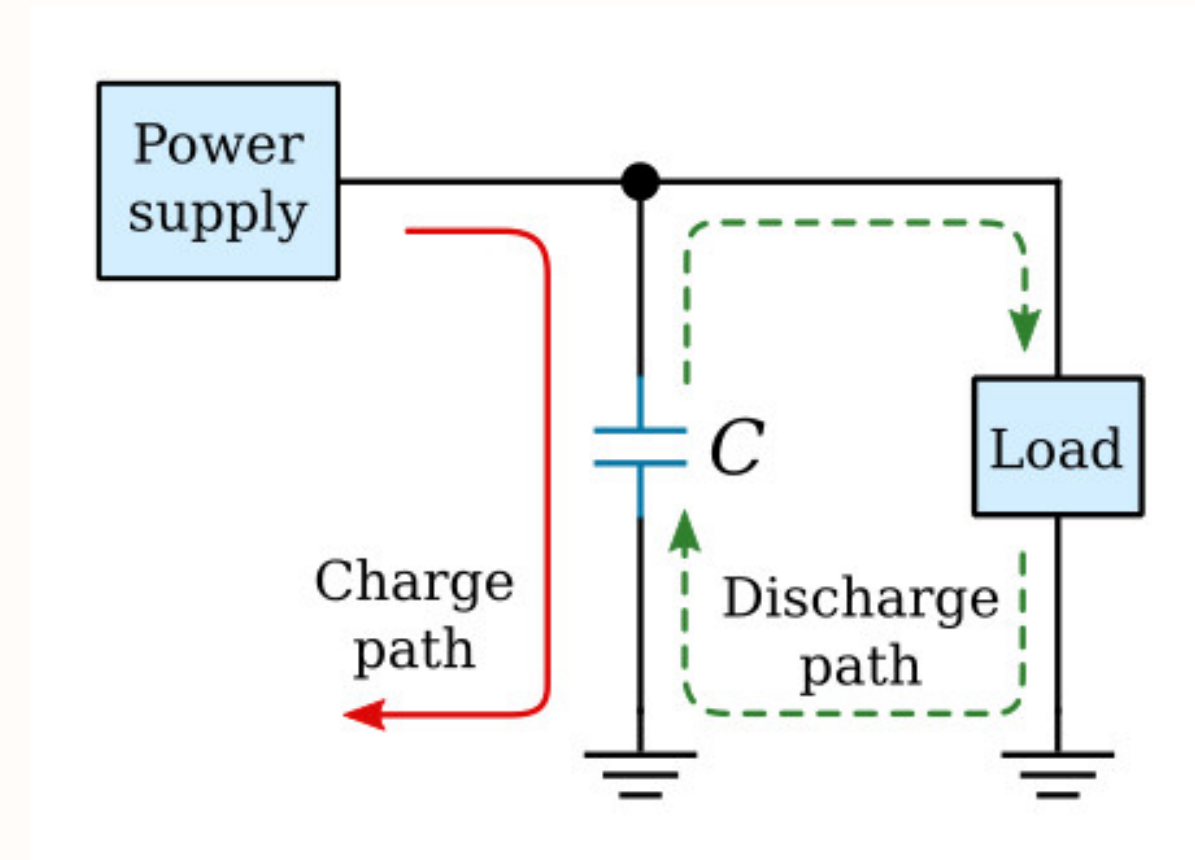
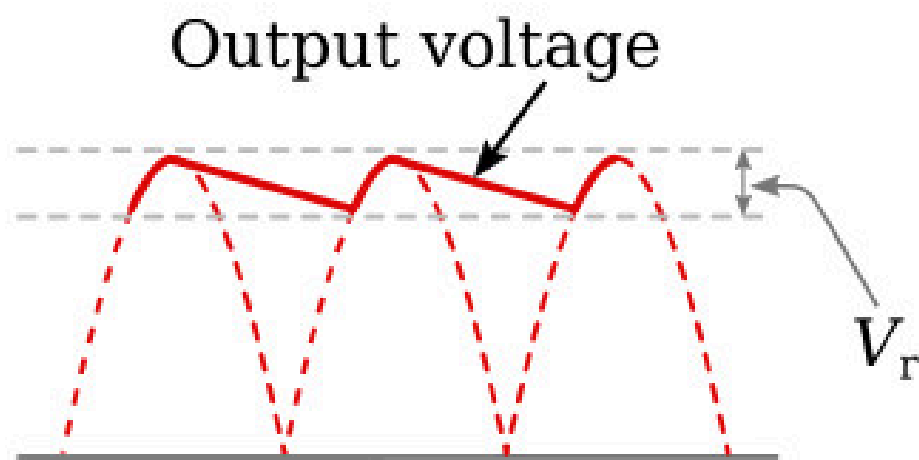
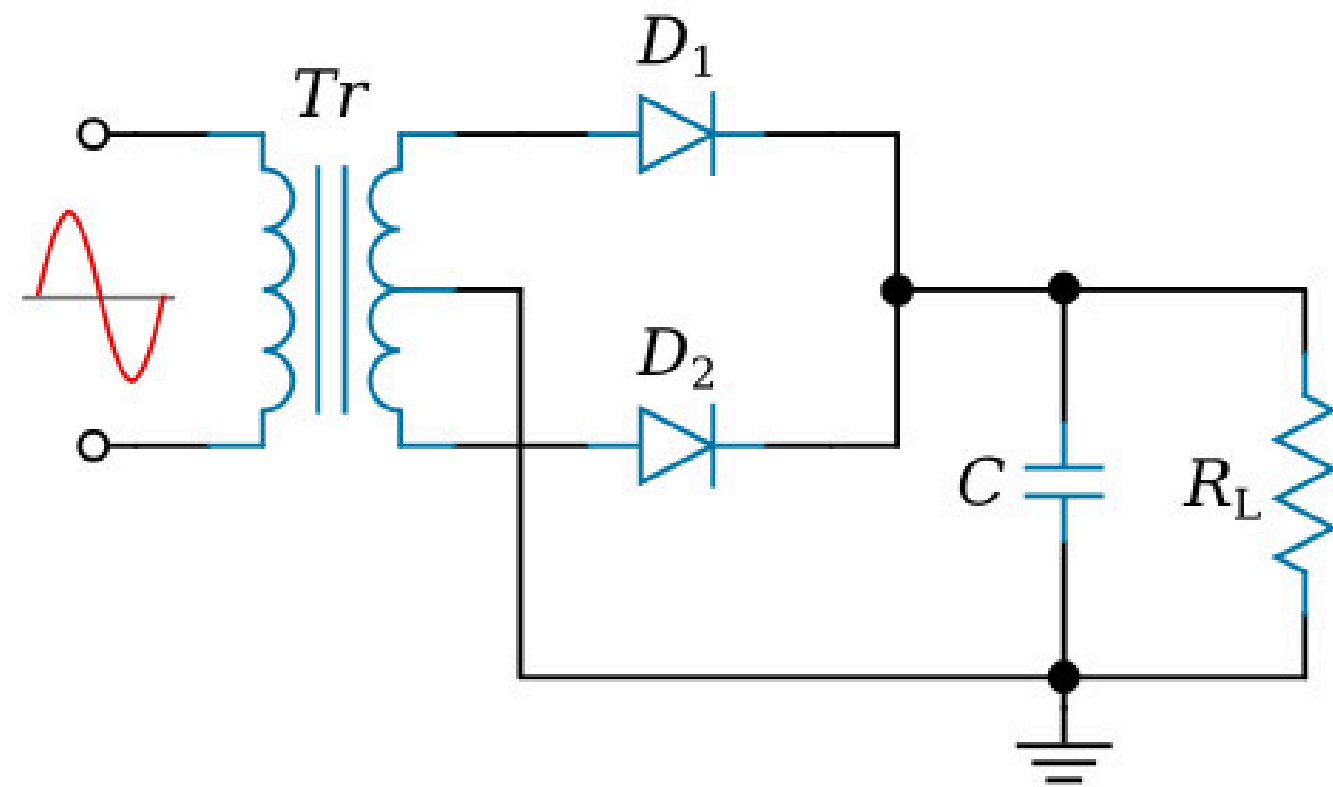
$$V_{AVG} = \frac{1}{\pi} \left[\int_0^\pi V_P \sin \theta d\theta \right] = \frac{V_P}{\pi} [-\cos \theta]_0^\pi = \frac{V_P}{\pi} (2) = 0.636 V_P$$

$$V_{AVG} = 0.636 \times V_P$$



Rectifier Circuit

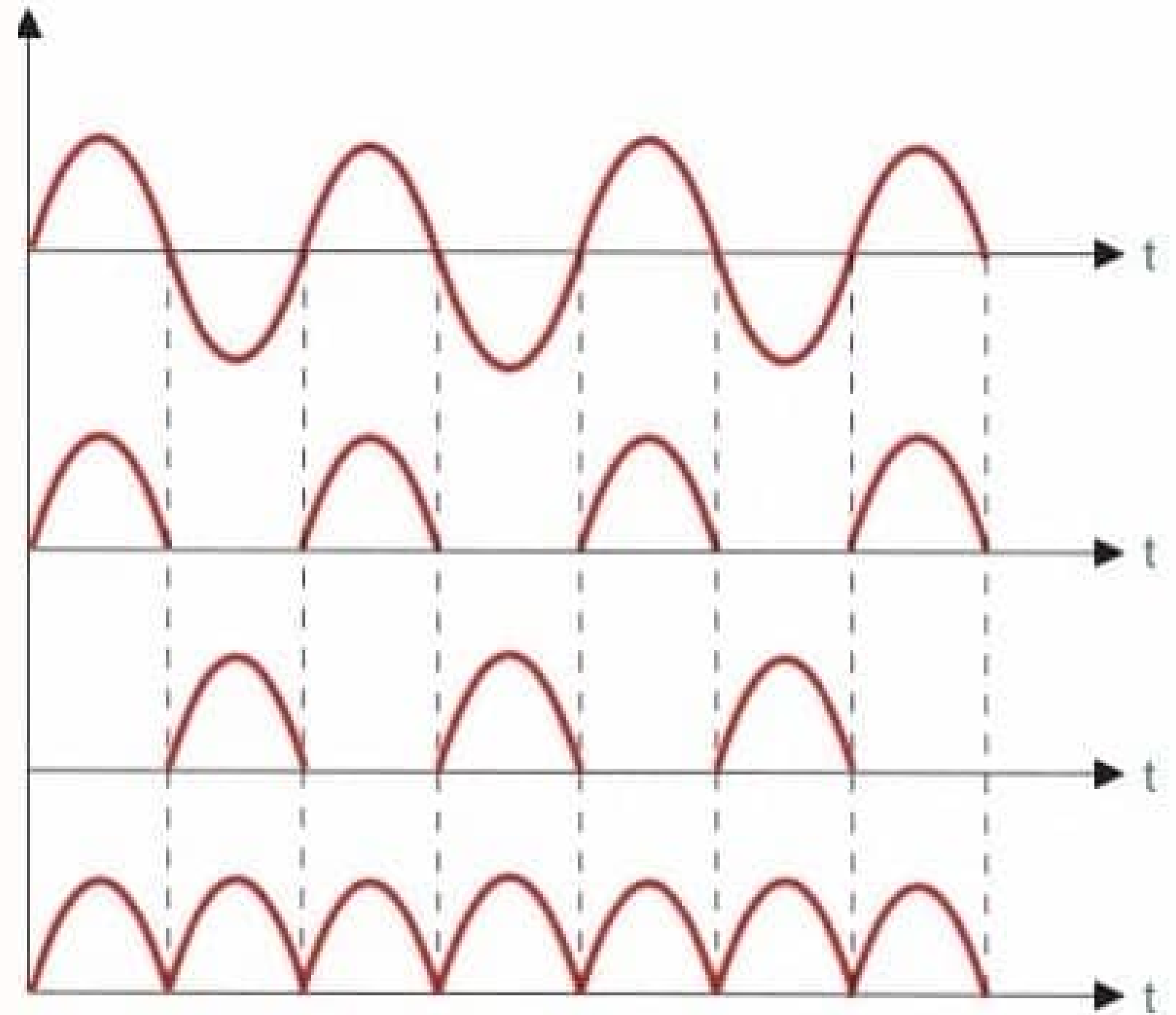
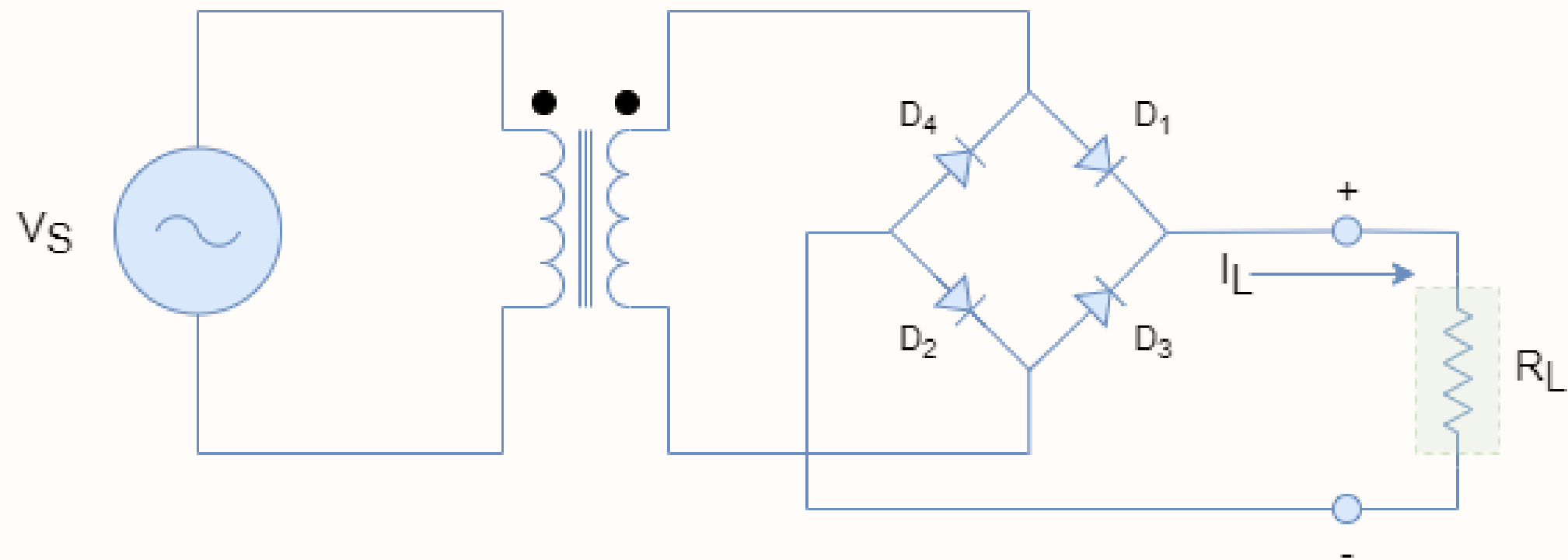
Full-wave rectifier (center-tapped)





Rectifier Circuit

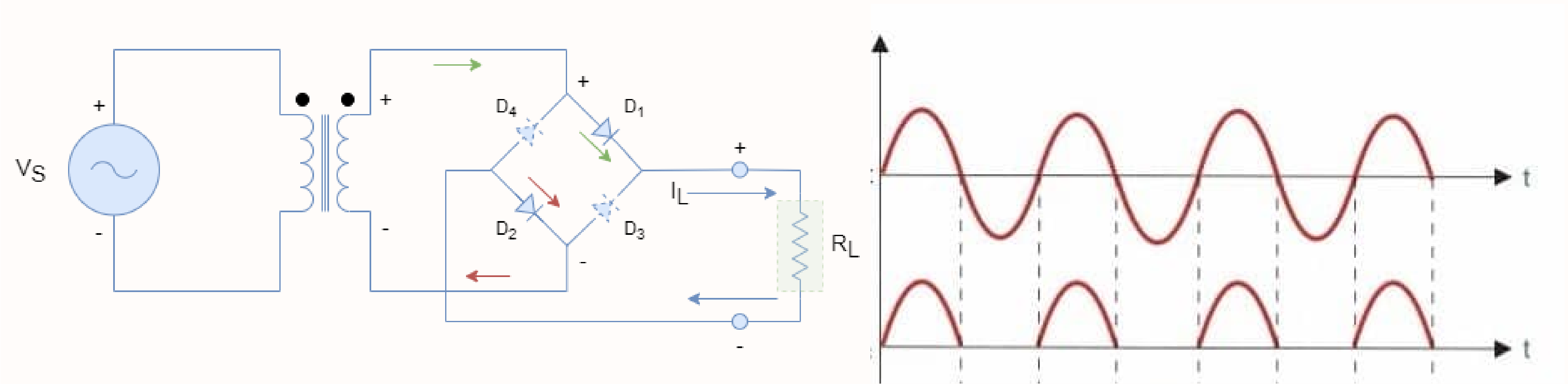
Full-wave rectifier (bridge)





Rectifier Circuit

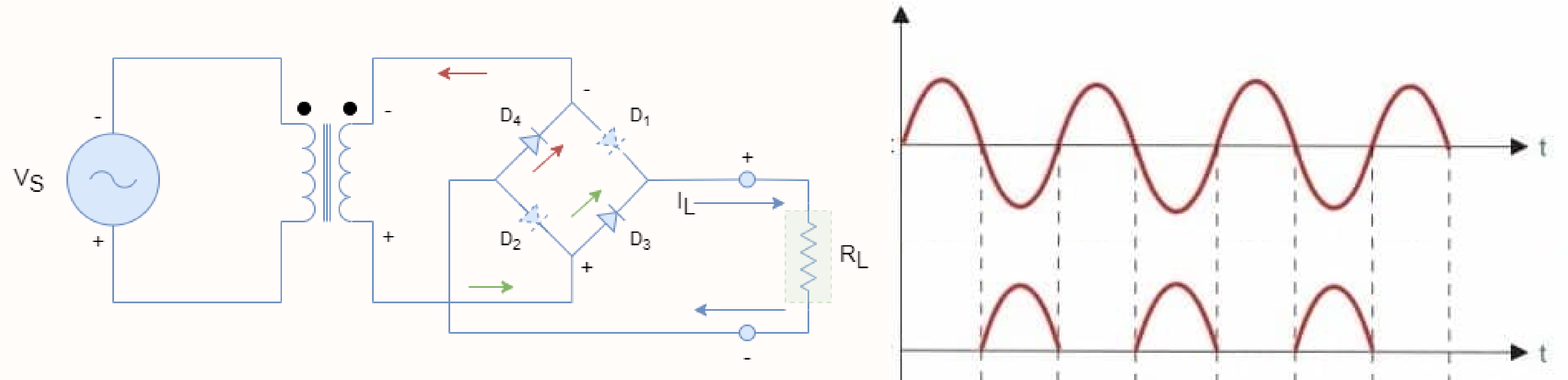
Full-wave rectifier (bridge)





Rectifier Circuit

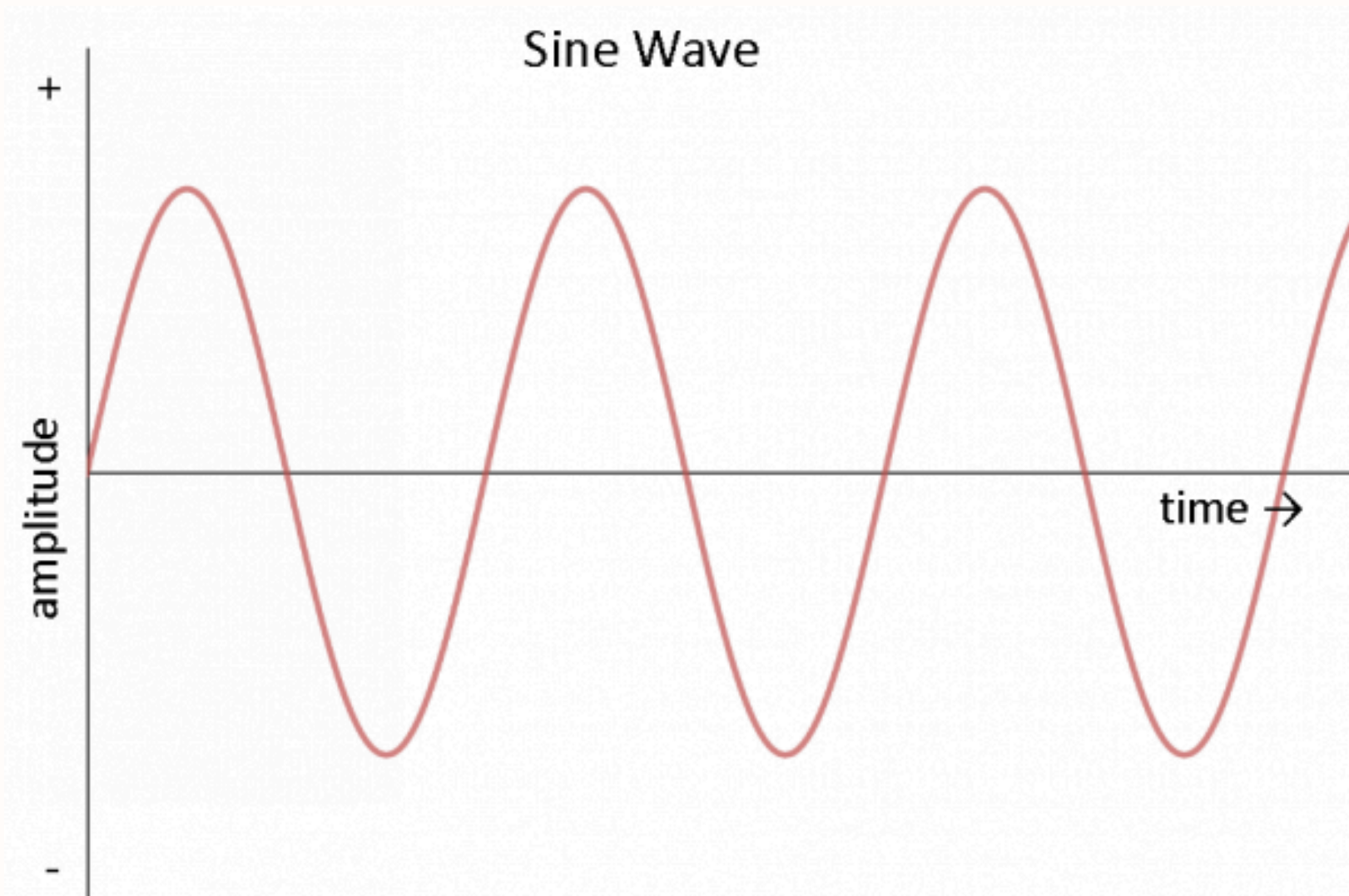
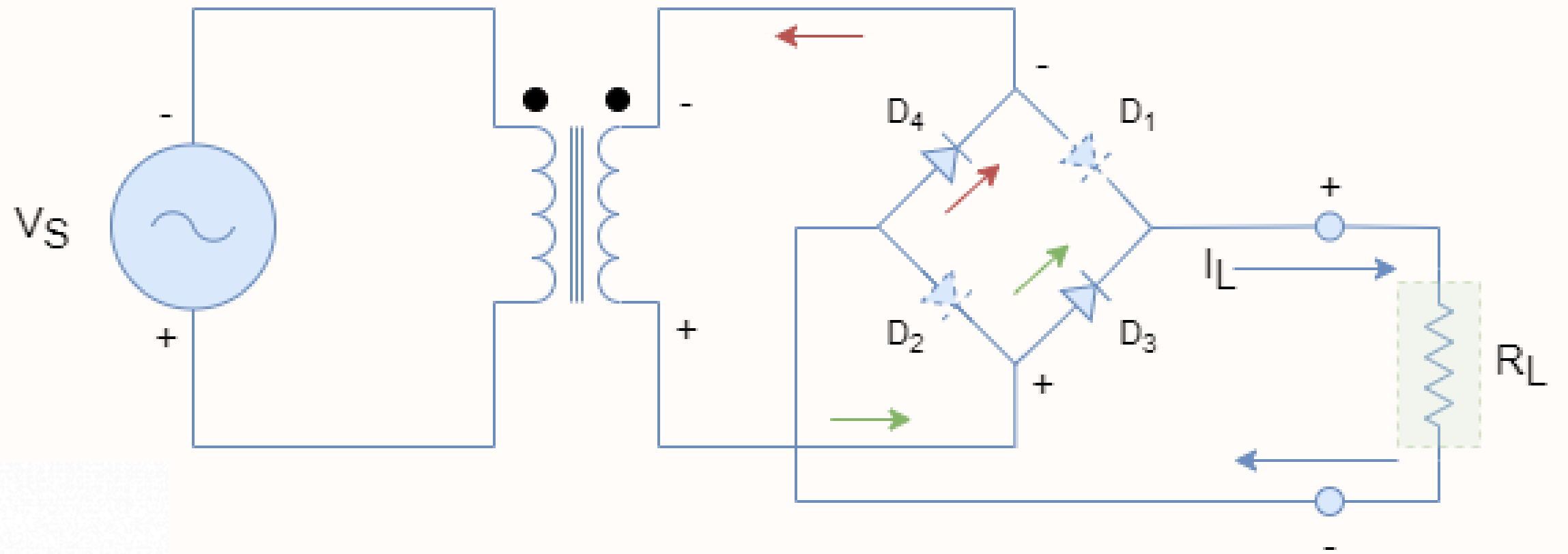
Full-wave rectifier (bridge)





Rectifier Circuit

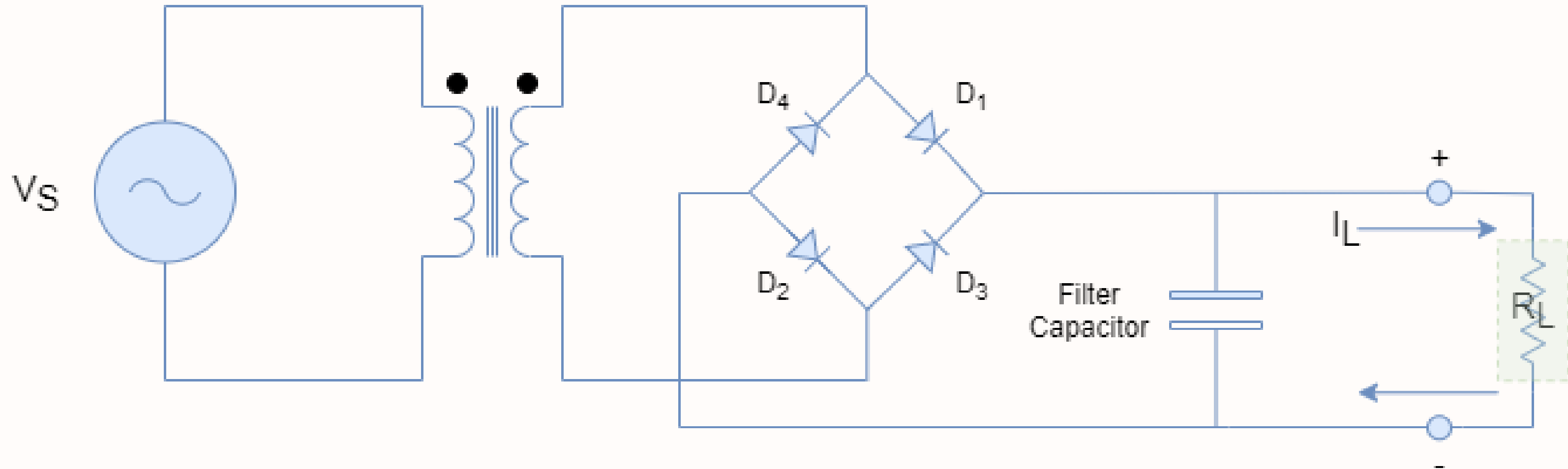
Bridge rectifier





Rectifier Circuit

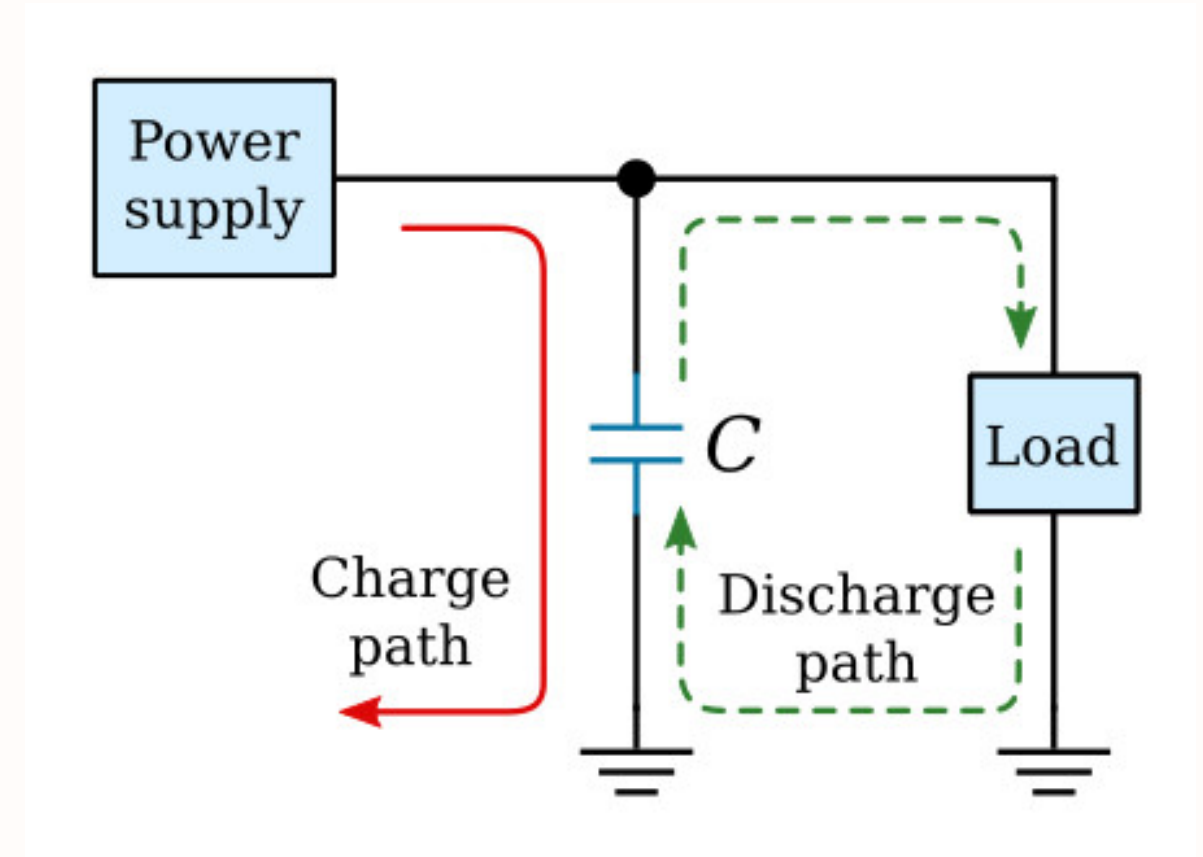
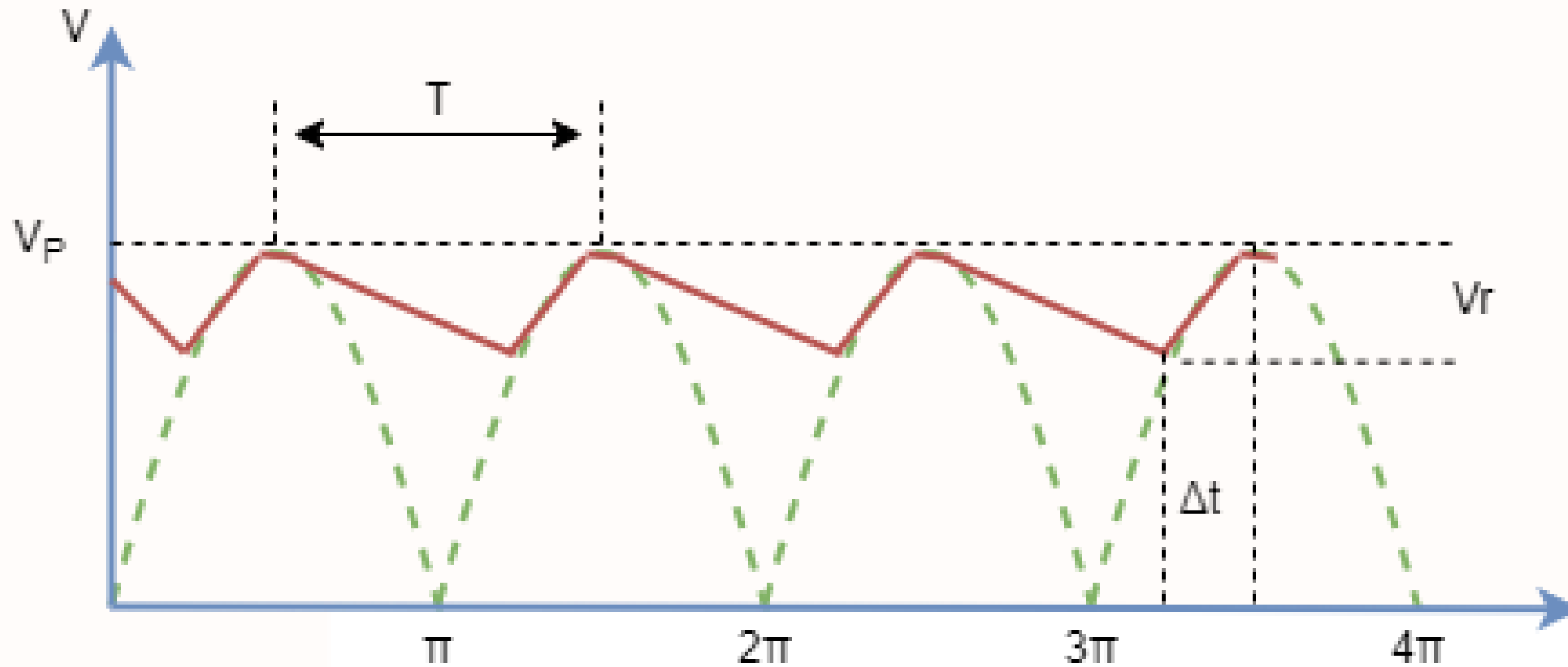
Bridge rectifier





Rectifier Circuit

Bridge rectifier





Rectifier Circuit

Half-wave rectifier applications (low power):

- Rectification applications
- Signal demodulation purpose
- Signal peak applications

Advantages	Disadvantages
Simple	Low output power and voltage
Cheap	Higher ripple factor
Easy to use	Low transformer utilization factor



Rectifier Circuit

Full-wave rectifier applications:

- **They are used for identifying the amplitude of modulating radio signal**
- **Full wave bridge rectifiers are used in a power supply circuit for different applications.**
- **They are used to provide power supply to devices that require DC voltage similar to LED and motor.**
- **They are used in mobile phones, laptops, and charger circuits.**
- **They are used in UPS to convert AC into DC.**
- **They are used in welding.**



Rectifier Circuit

Full-wave rectifier

Advantages

higher efficiency

no voltage signal is wasted in the rectification process

less ripple factor

Disadvantages

more circuit elements

complex and tends to occupy a lot of space.

more expensive



Rectifier Circuit

Rectifier design

Step 1: Calculating the Load

Step 2: Selecting The Right Diode

Step 3: The Output Capacitor Filter

Step 4: Testing The Design Circuit



Rectifier Circuit

Rectifier design

We are given a load of wattage 0.2W and voltage rating 5V. The problem statement is, we need to design a half wave rectifier to power the device up from the main power AC i.e 220V.



Rectifier Circuit

Rectifier design

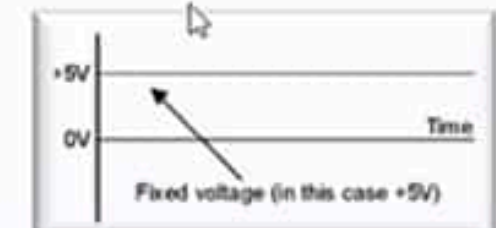
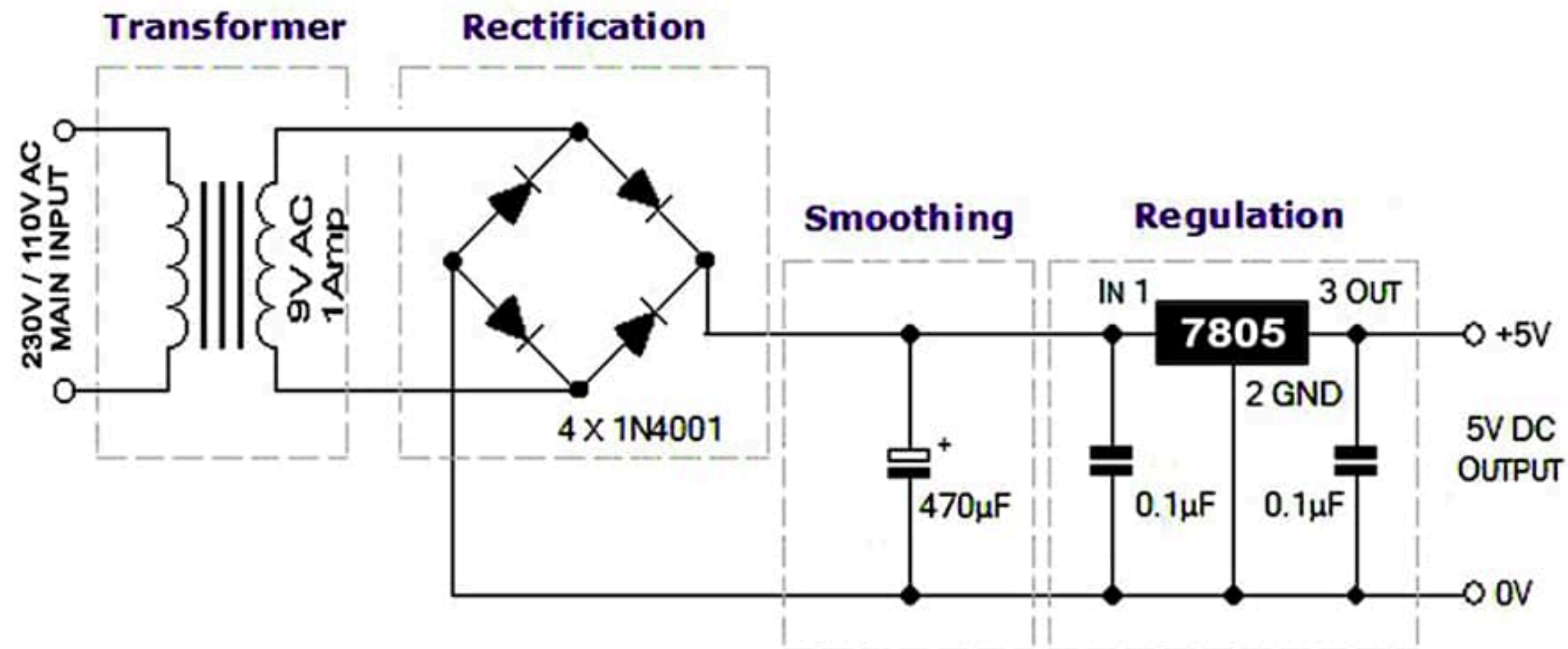
Design a rectifier/filter that will produce an output voltage of approximately 30 volts with a maximum current draw of 300 milliamps. It is to be fed from a 120 VAC RMS source. The ripple voltage should be less than 10% of the nominal output voltage at full load.



Rectifier Circuit

Simple DC power supply

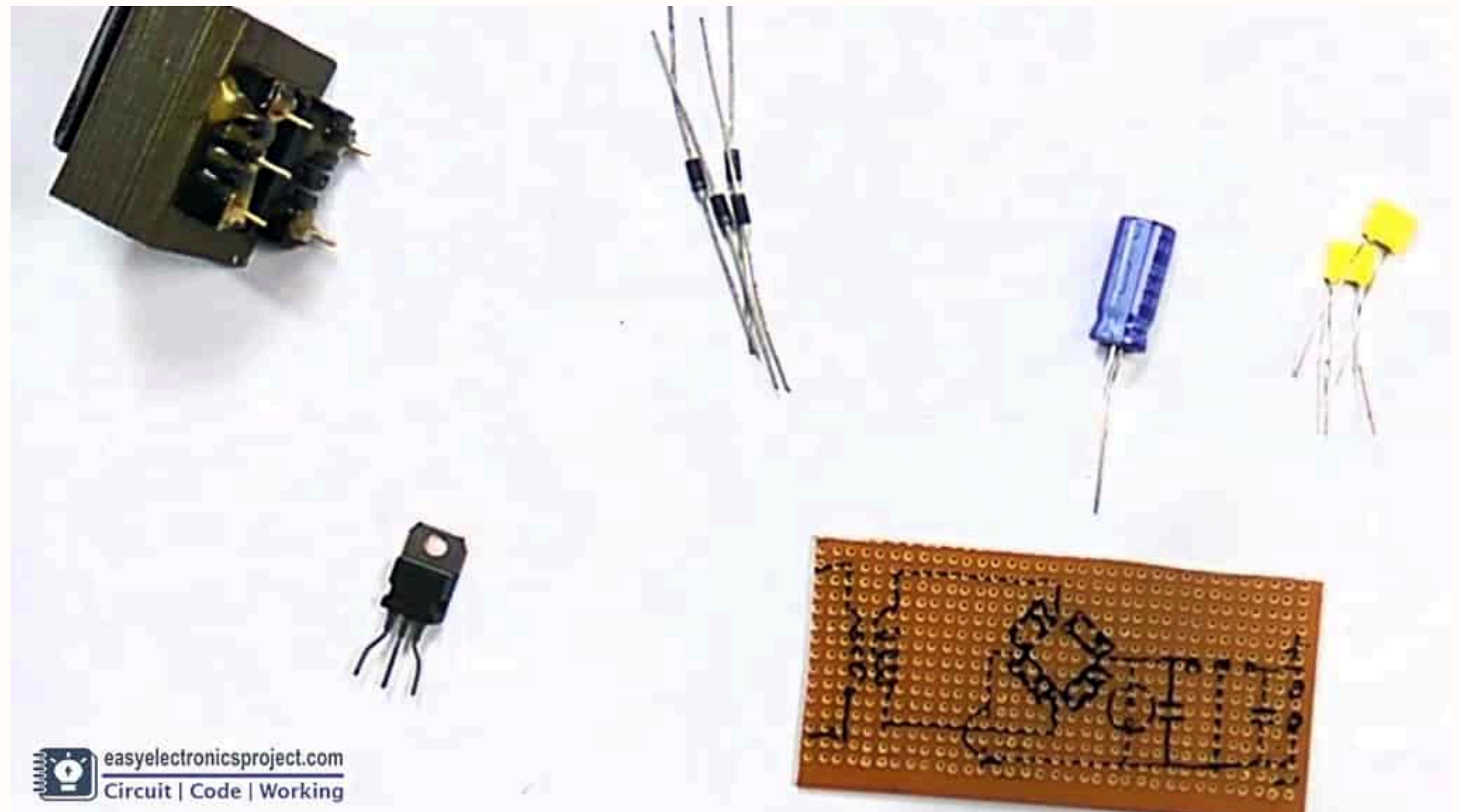
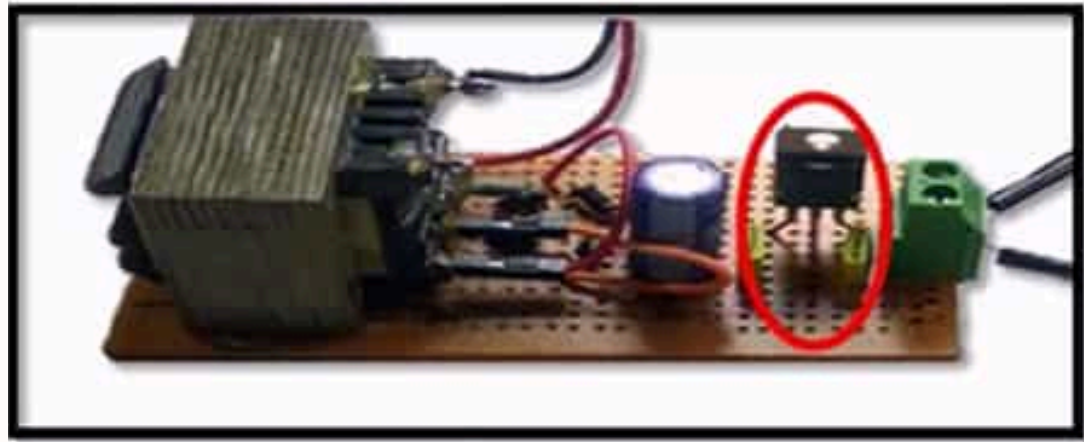
5 volt DC Power Supply using 7805





Rectifier Circuit

Simple DC power supply

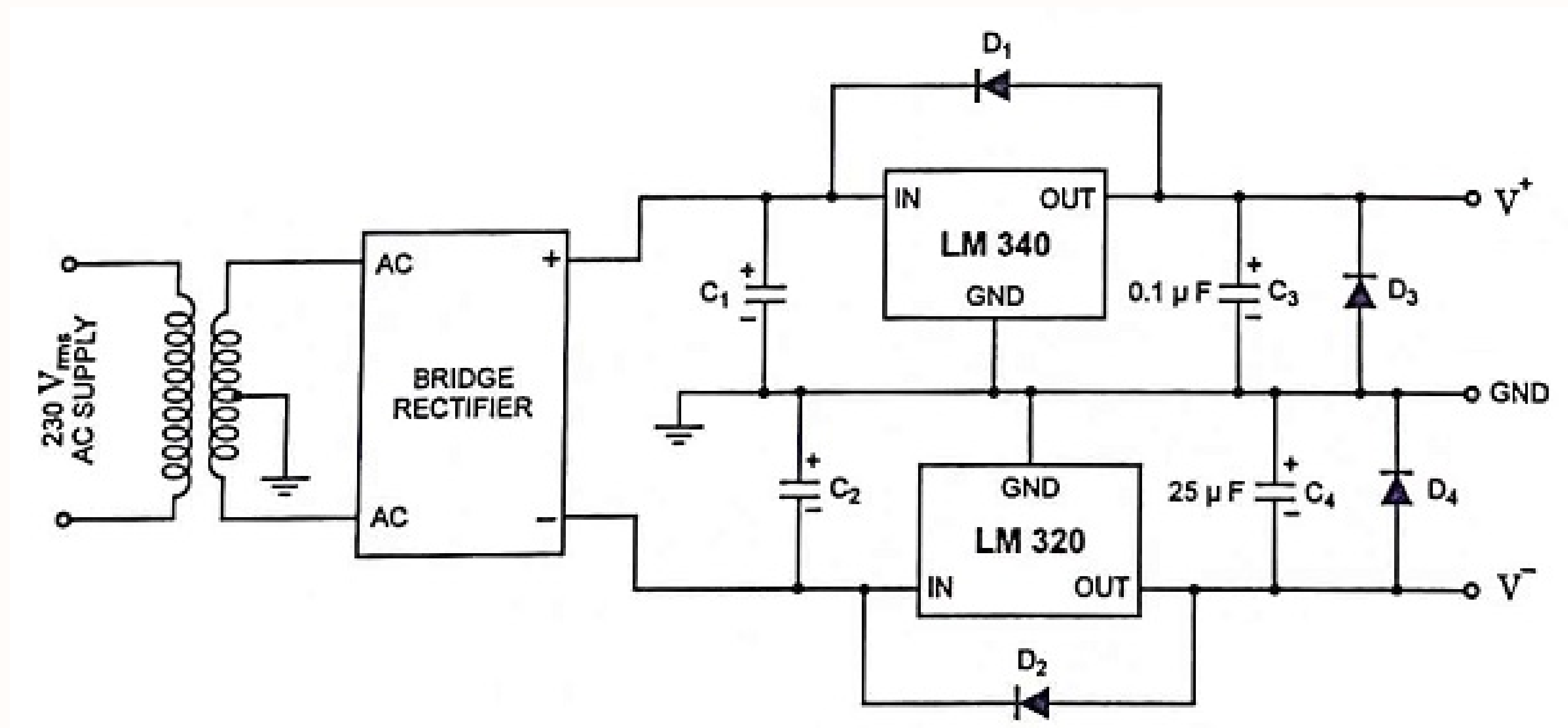


<https://youtu.be/1Gbh9NA6UOs?si=fcFeSy0jcPKY0GNd>



Rectifier Circuit

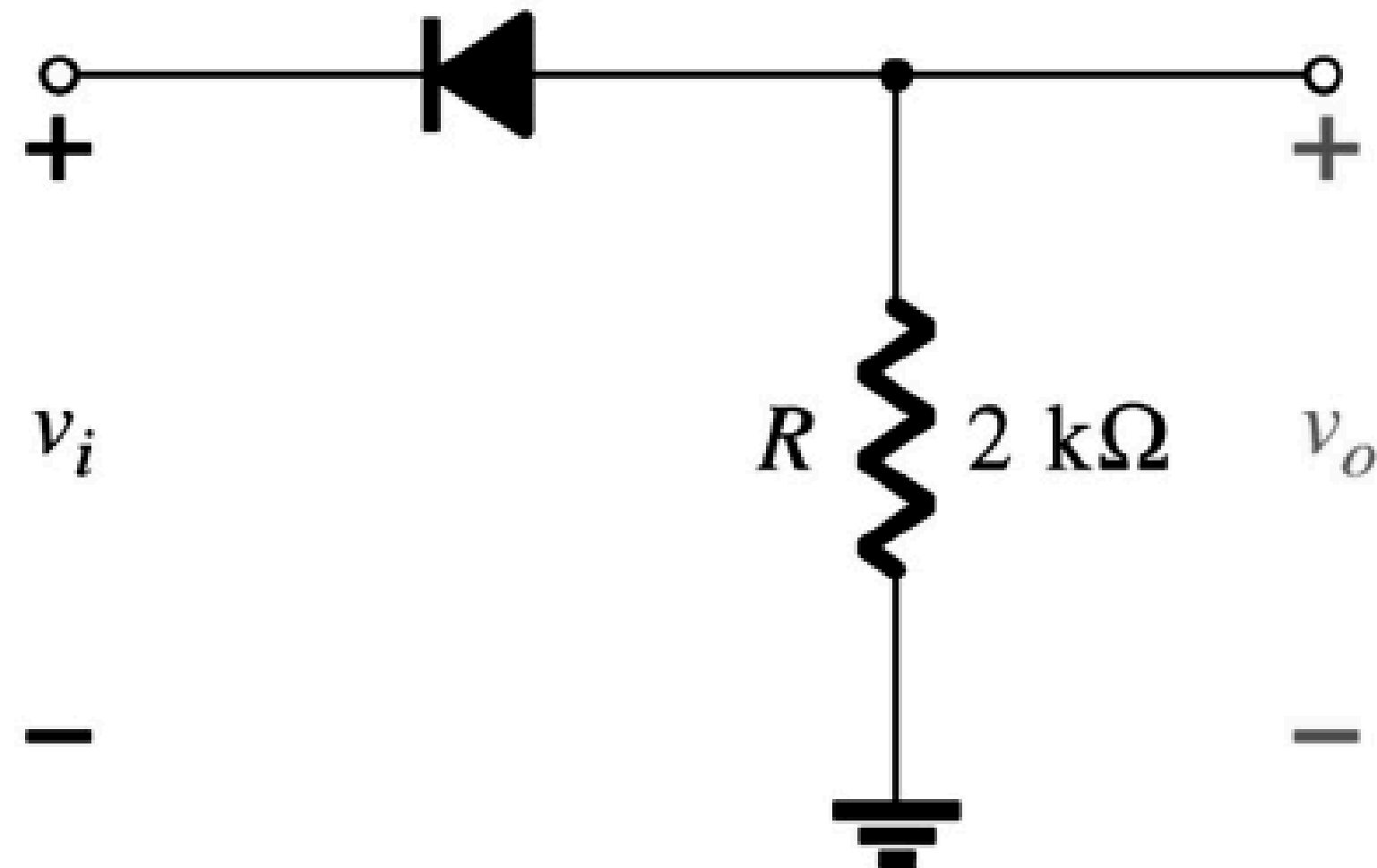
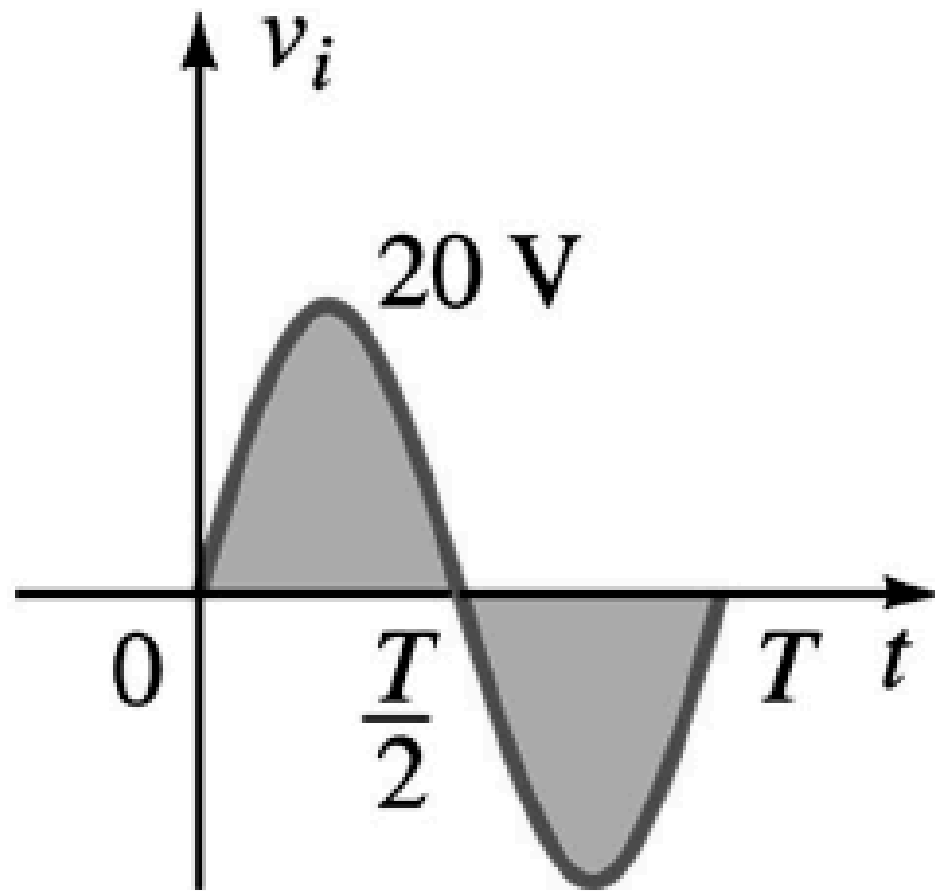
Dual power supply



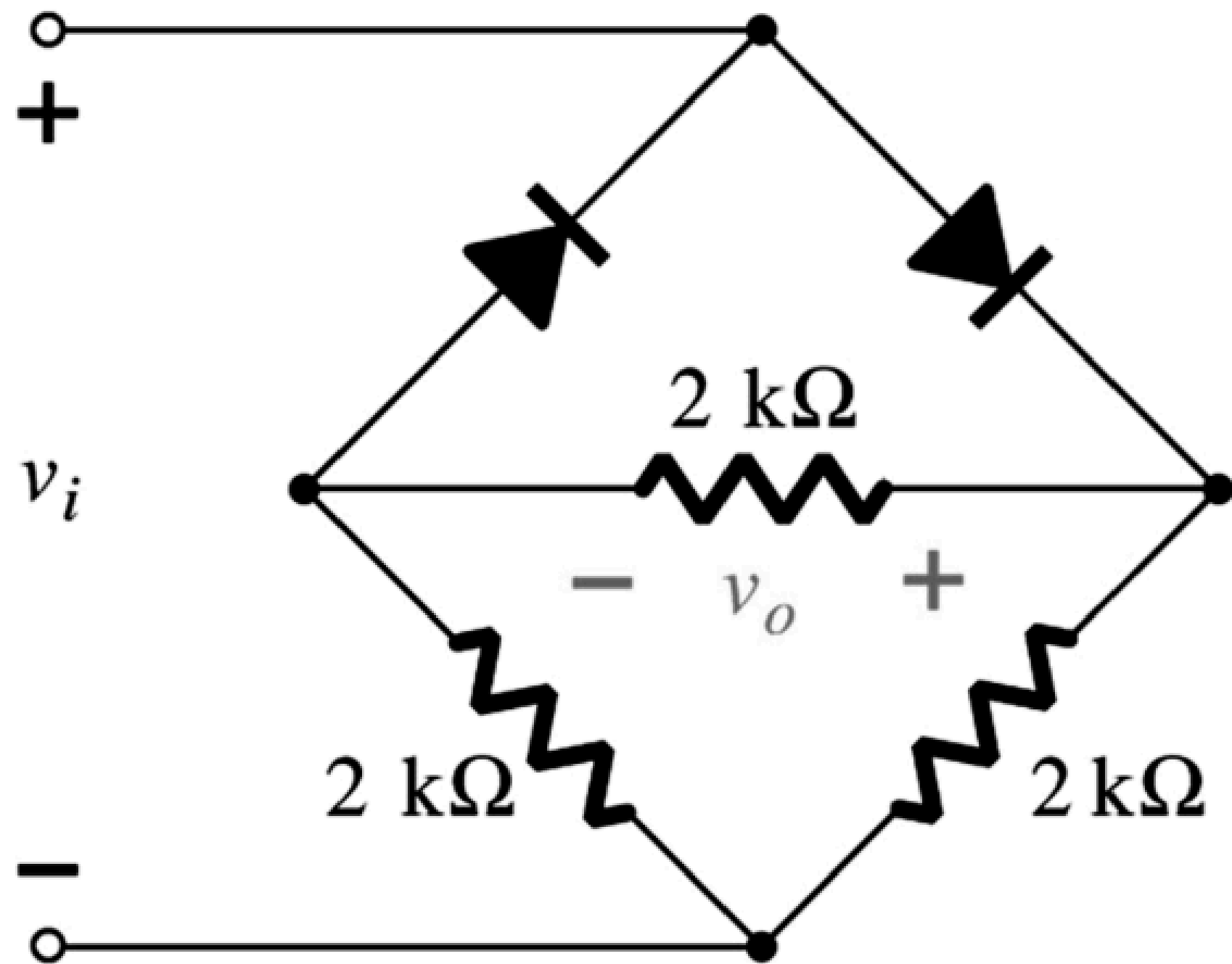
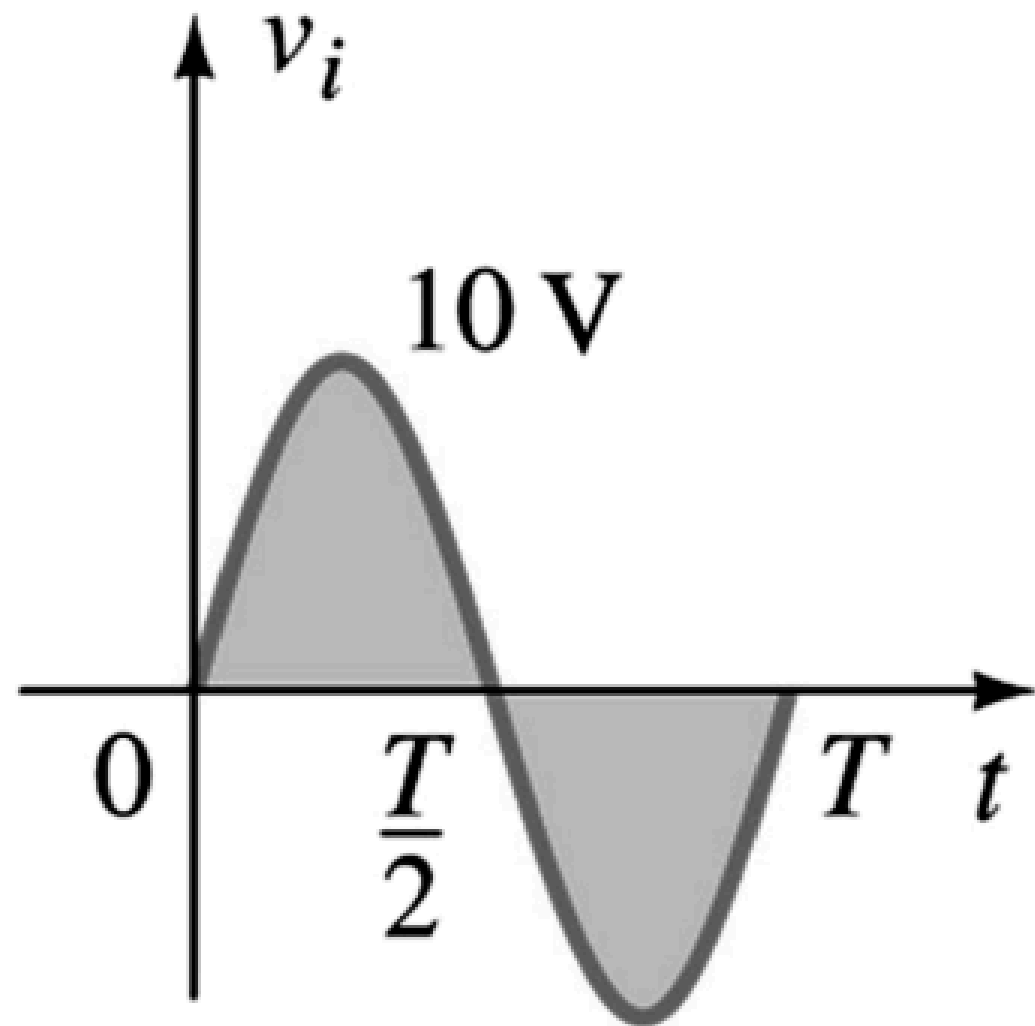


Rectifier Circuit

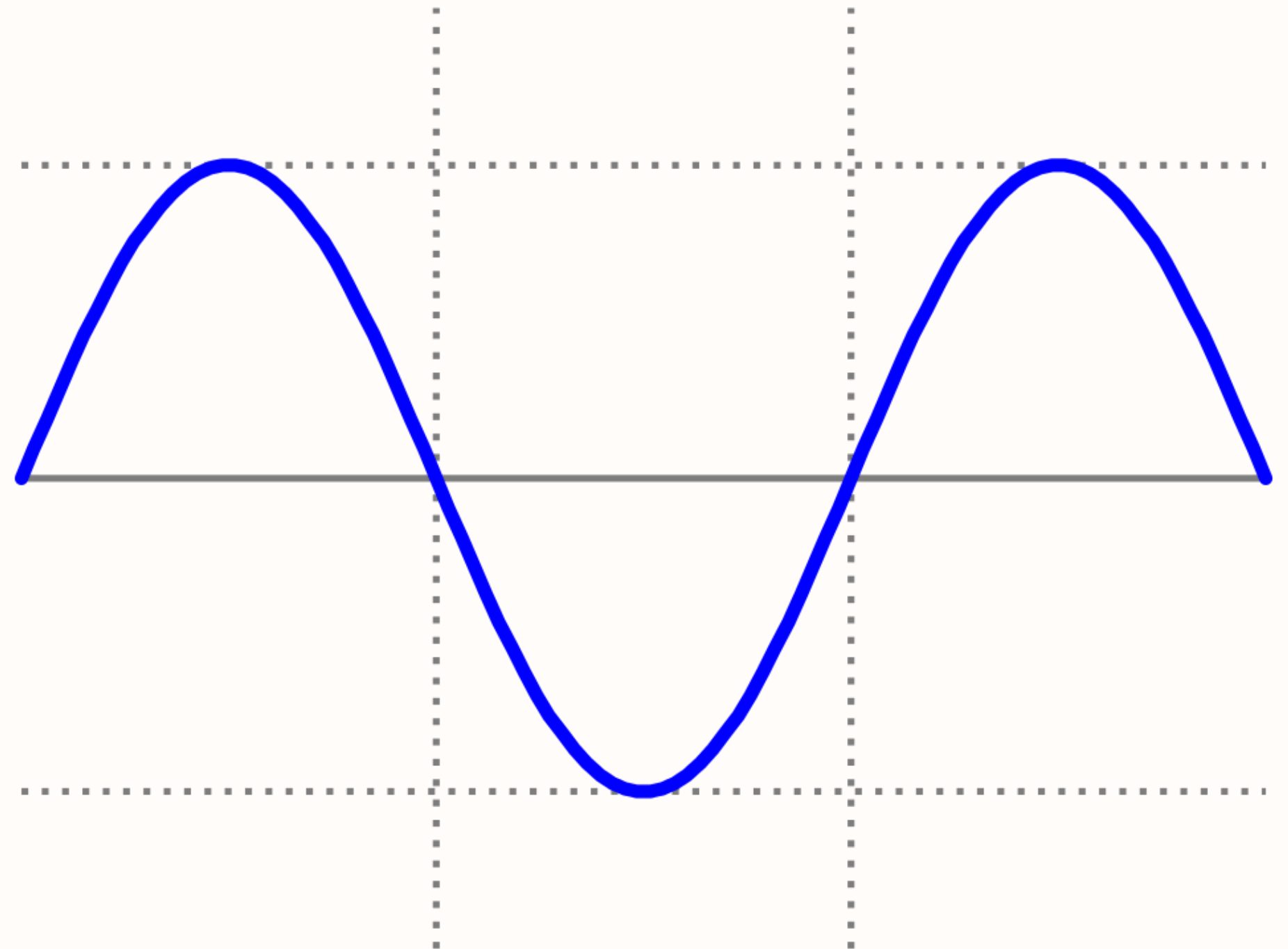
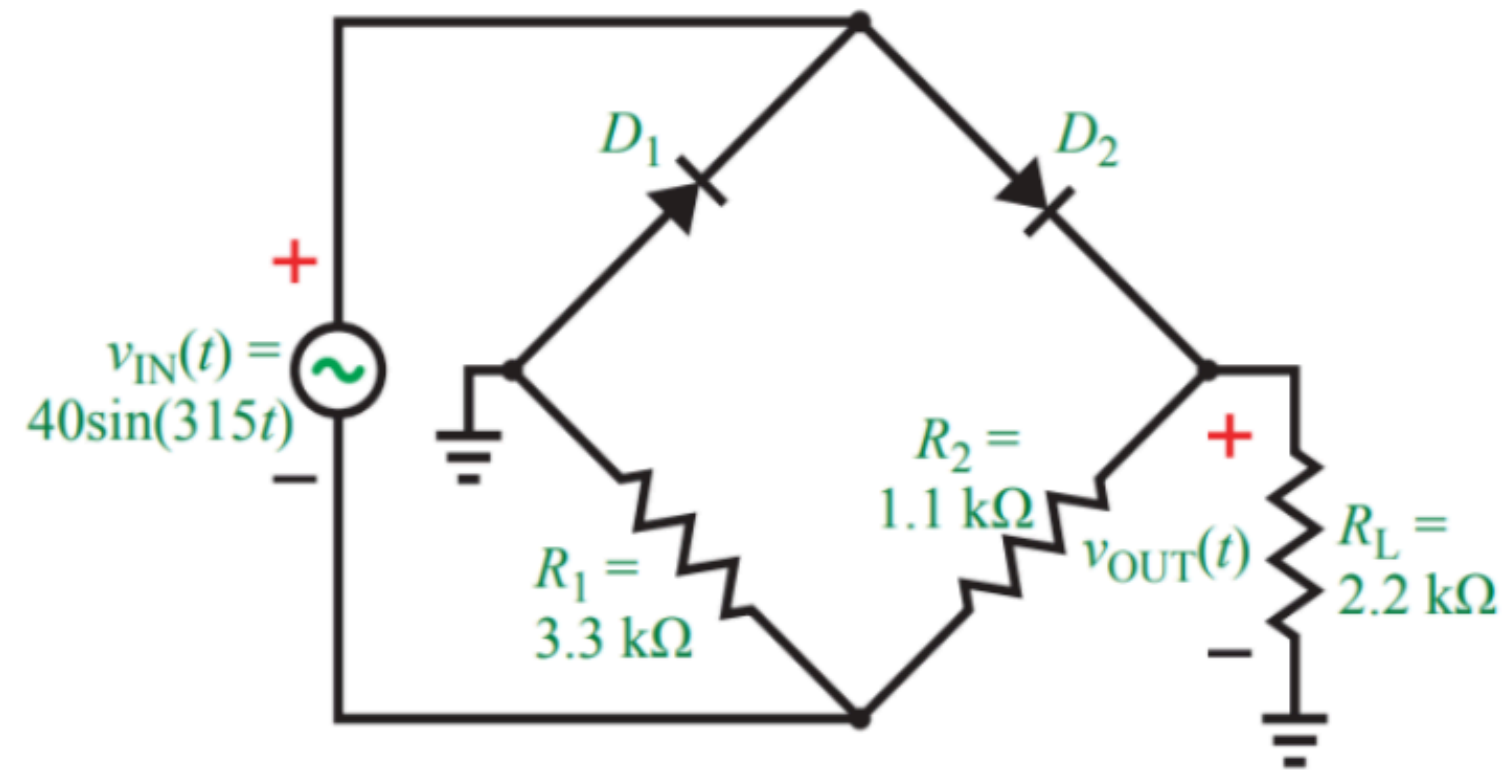
Let's think about it:



What is the output voltage waveform?

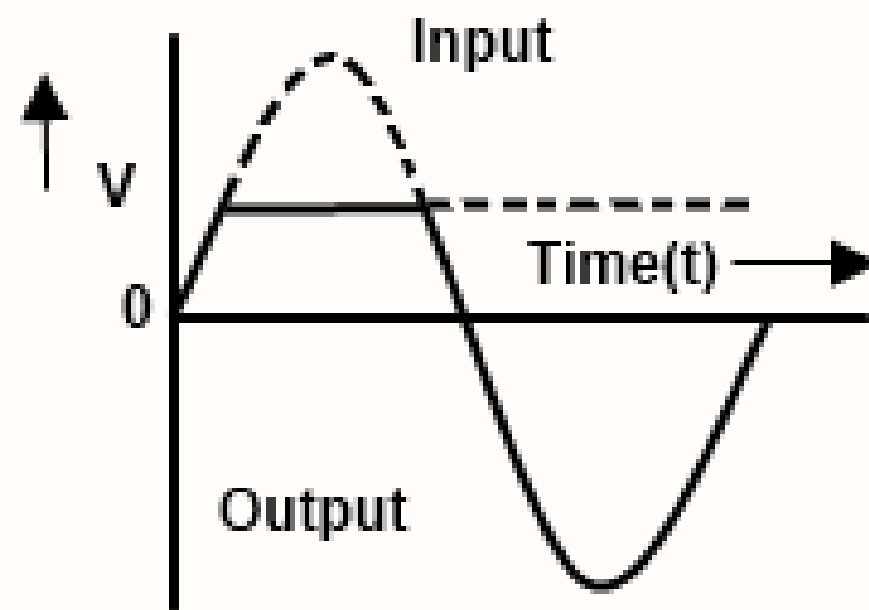
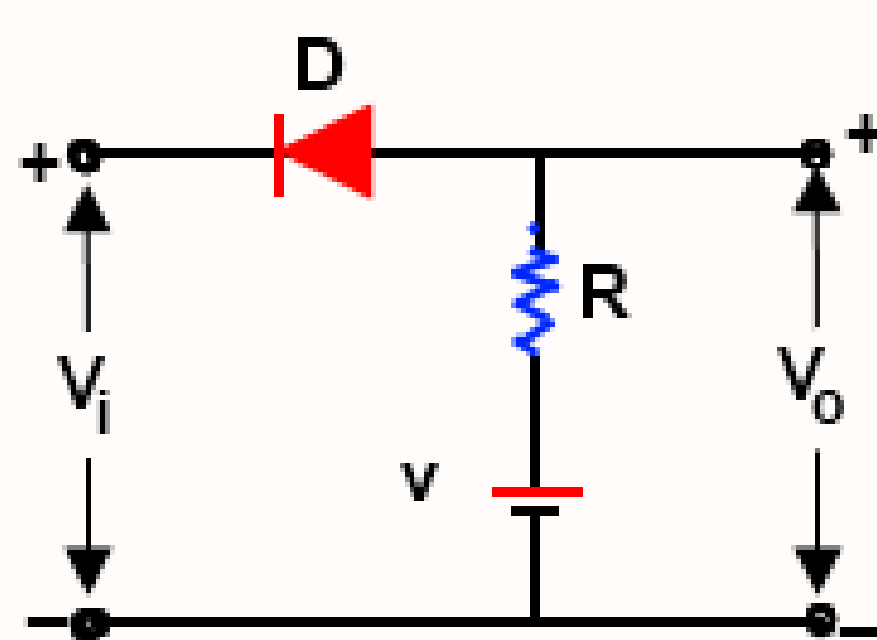


What is the output voltage waveform?

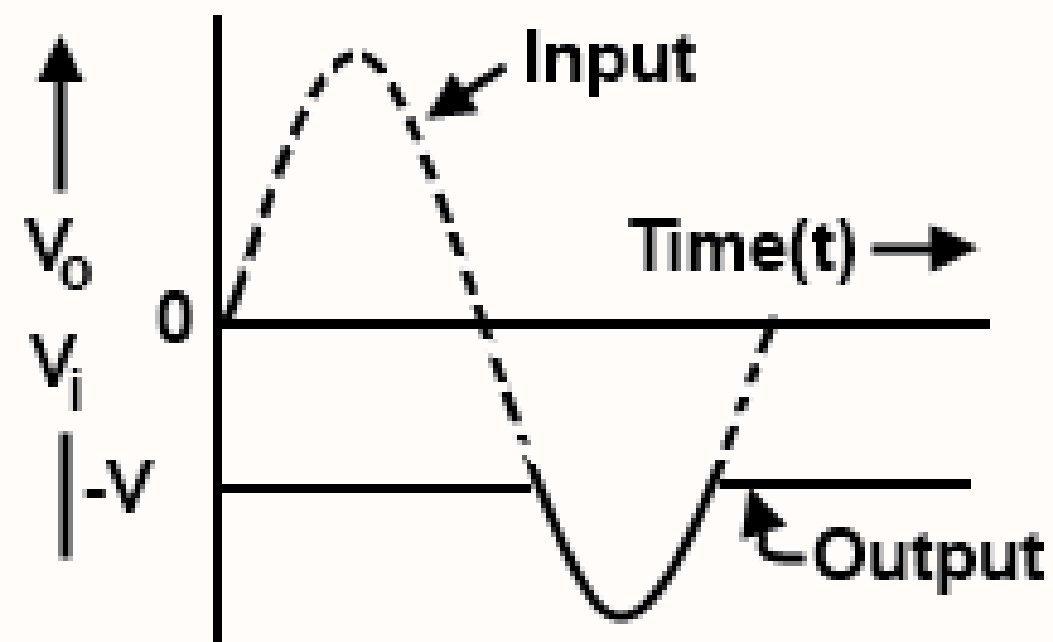
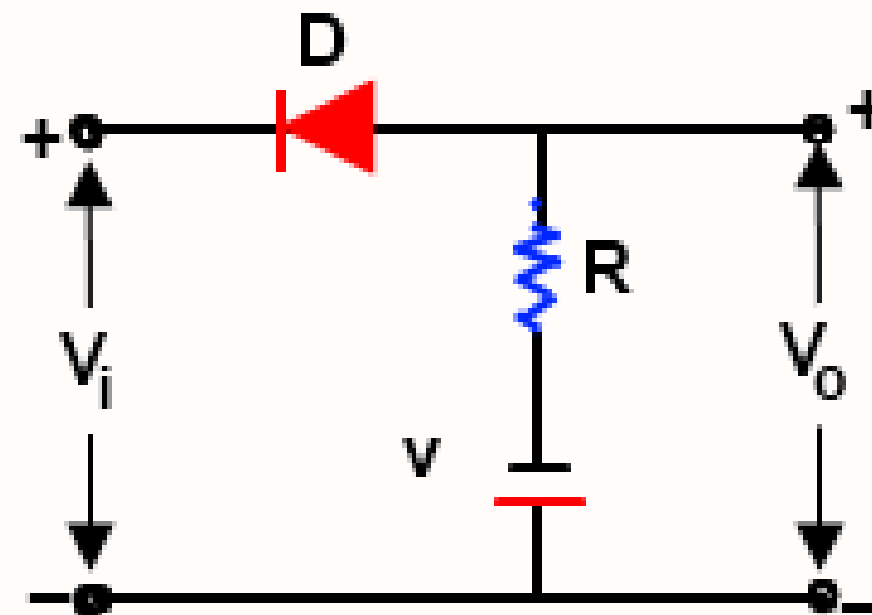




Diode Circuit



(a) Positive Biased

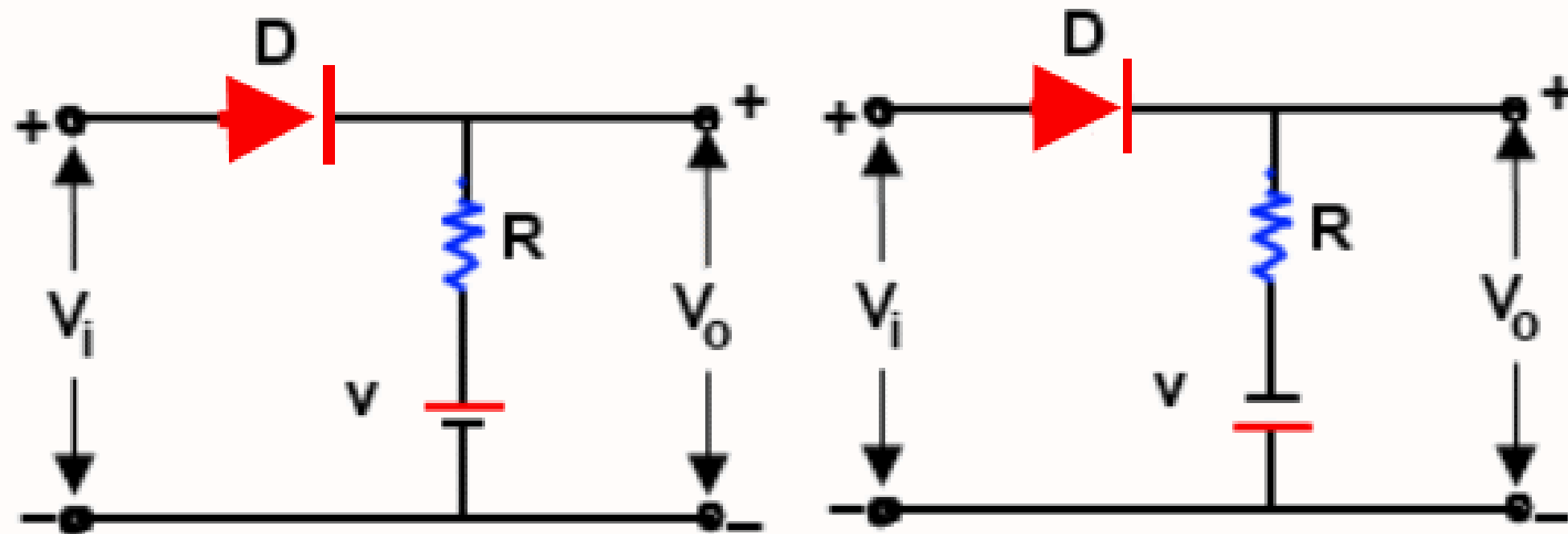


(b) Negative Biased

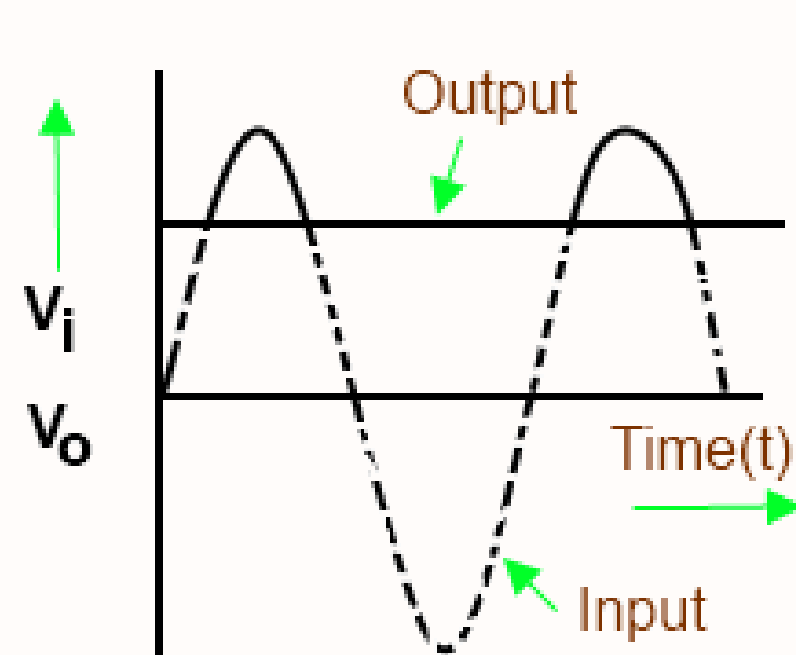
Clipper Circuits



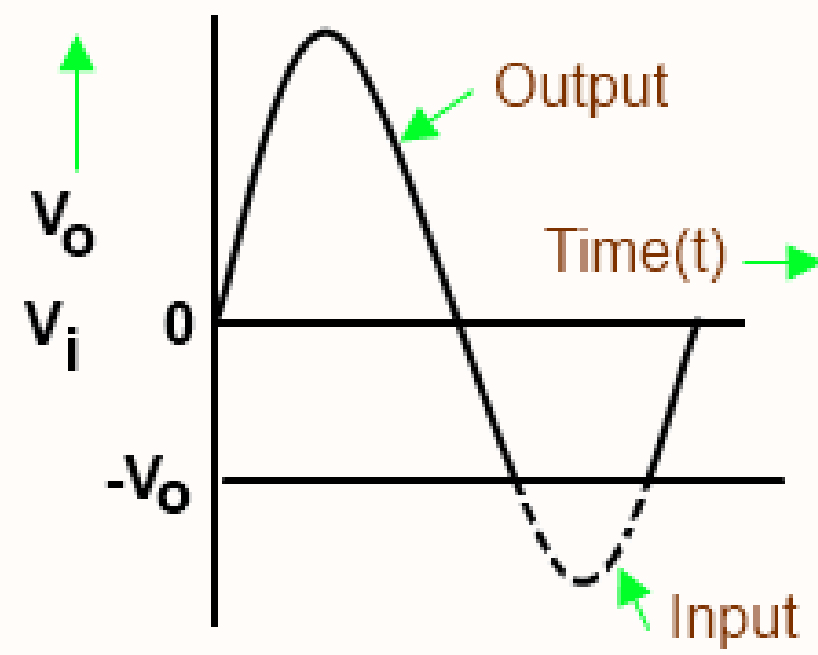
Diode Circuit



Clipper Circuits



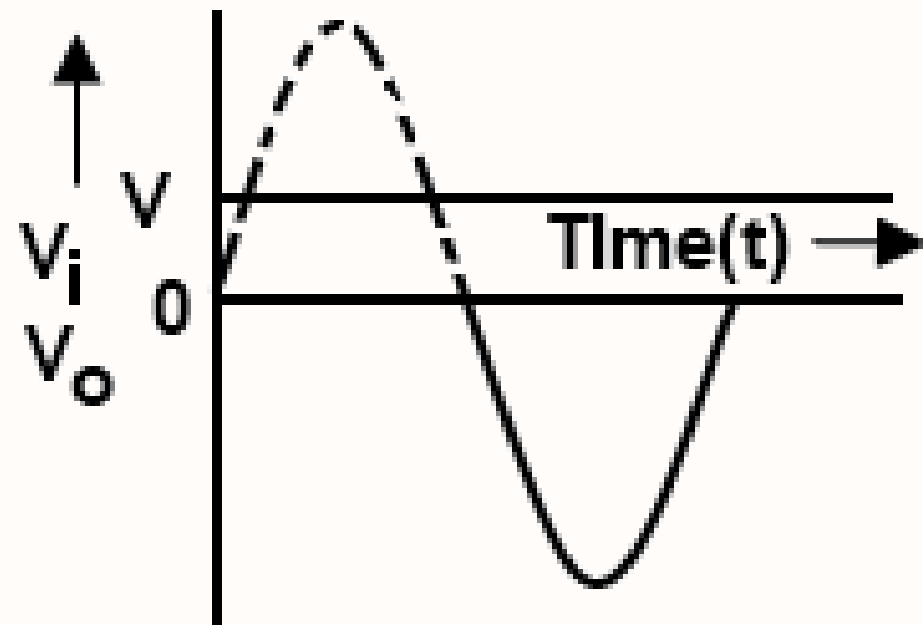
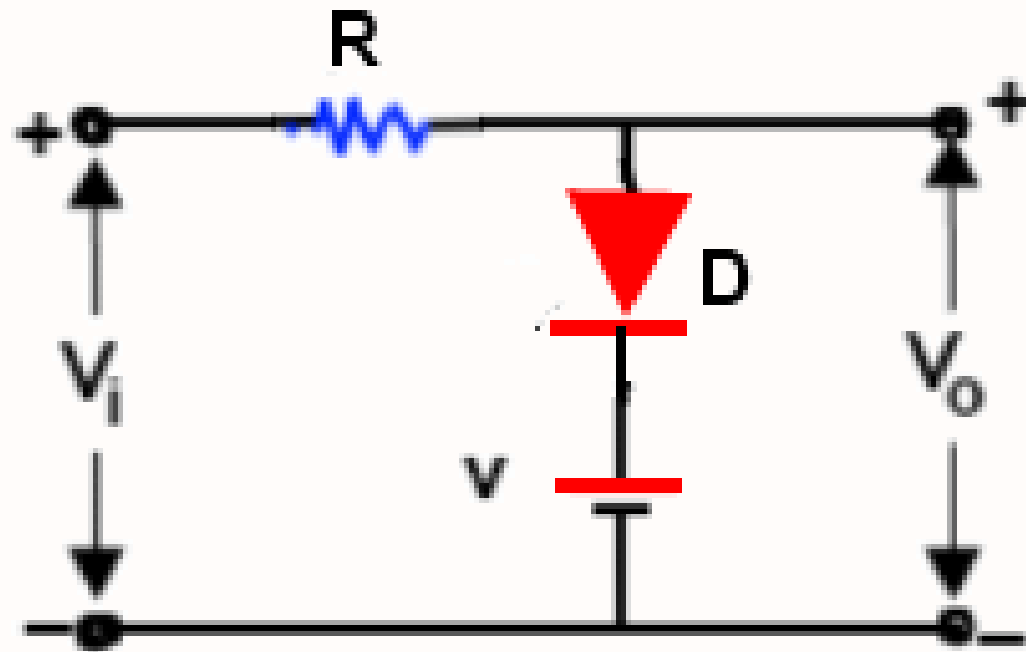
(a) Positive biased



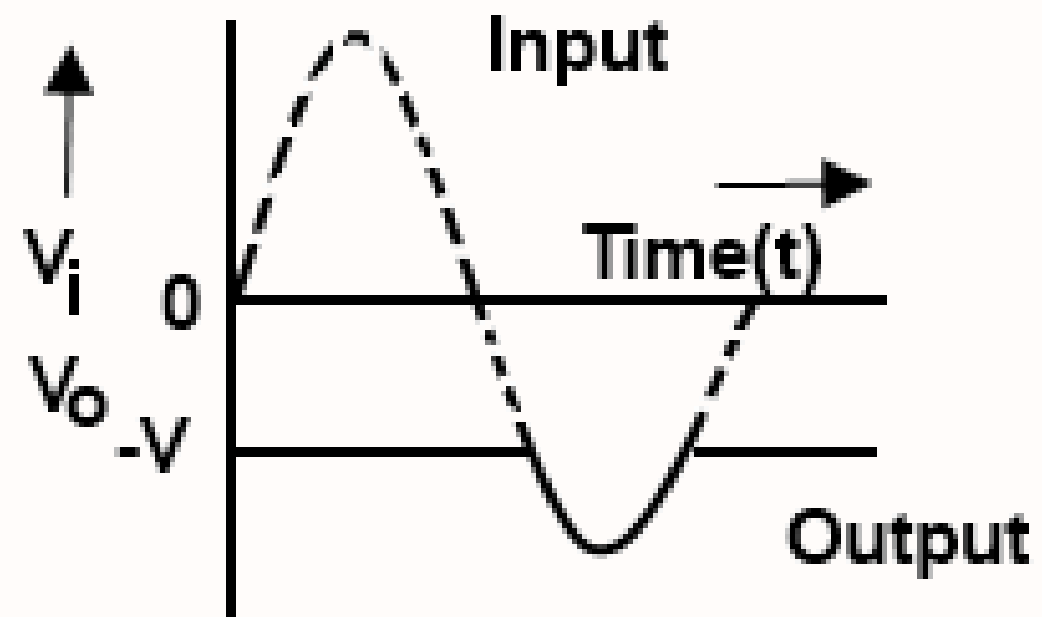
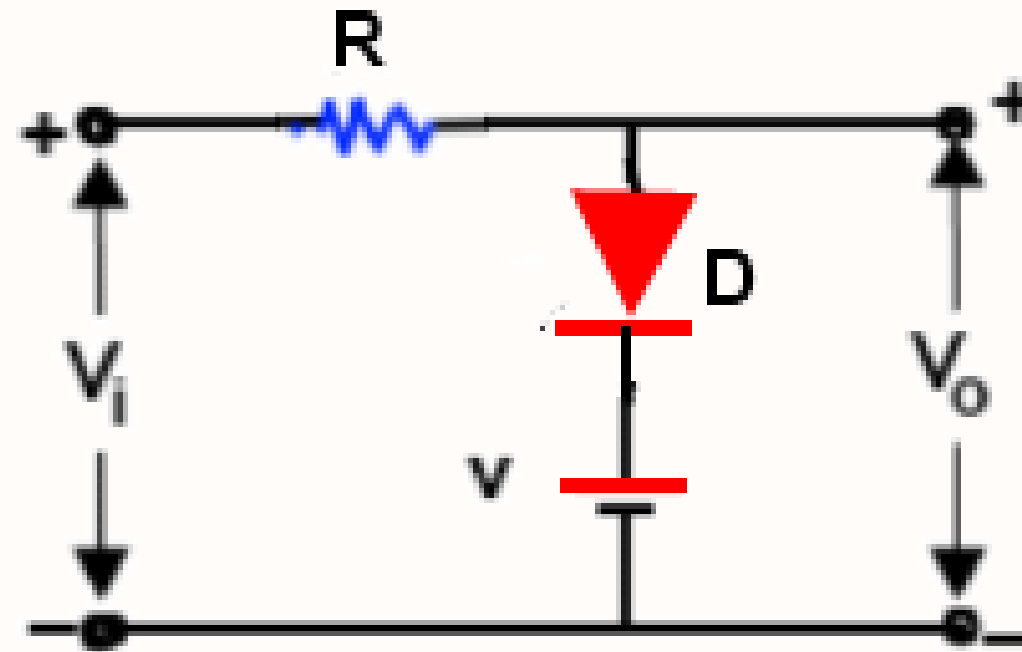
(b) Negative biased



Diode Circuit



(a) Positive biased

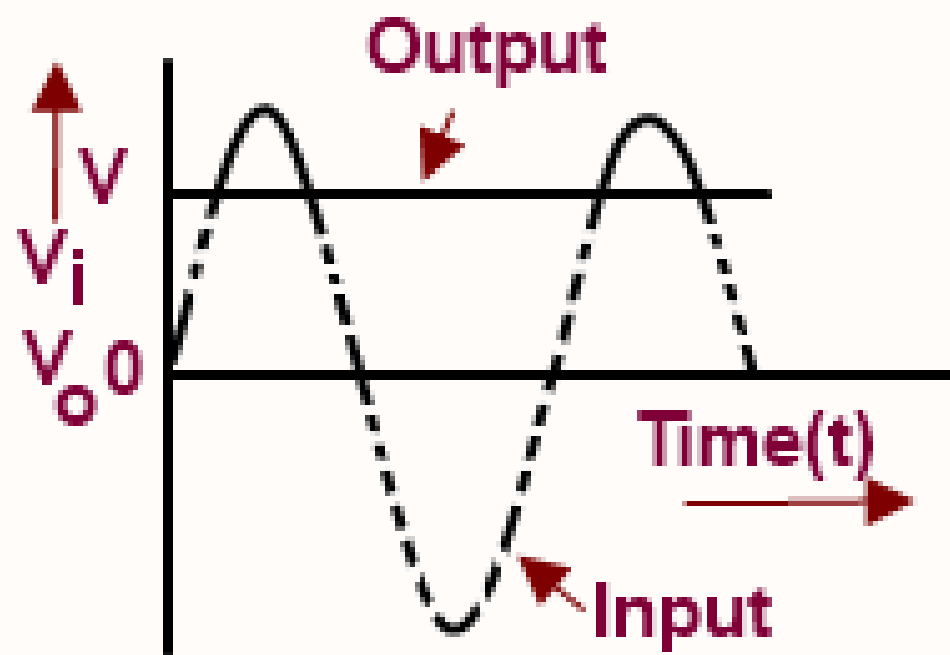
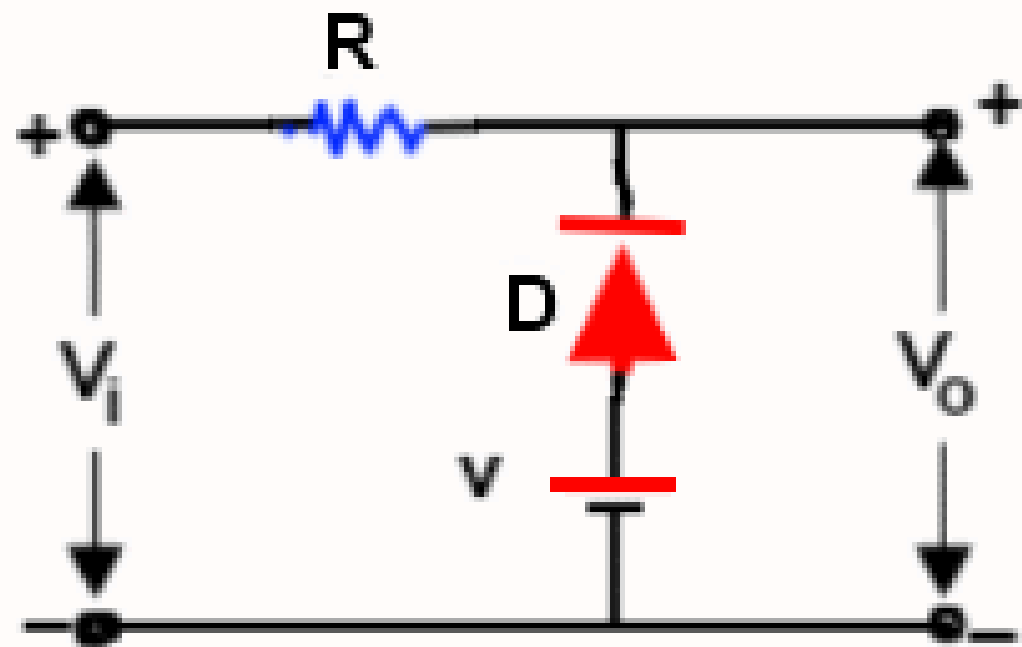


(b) Negative biased

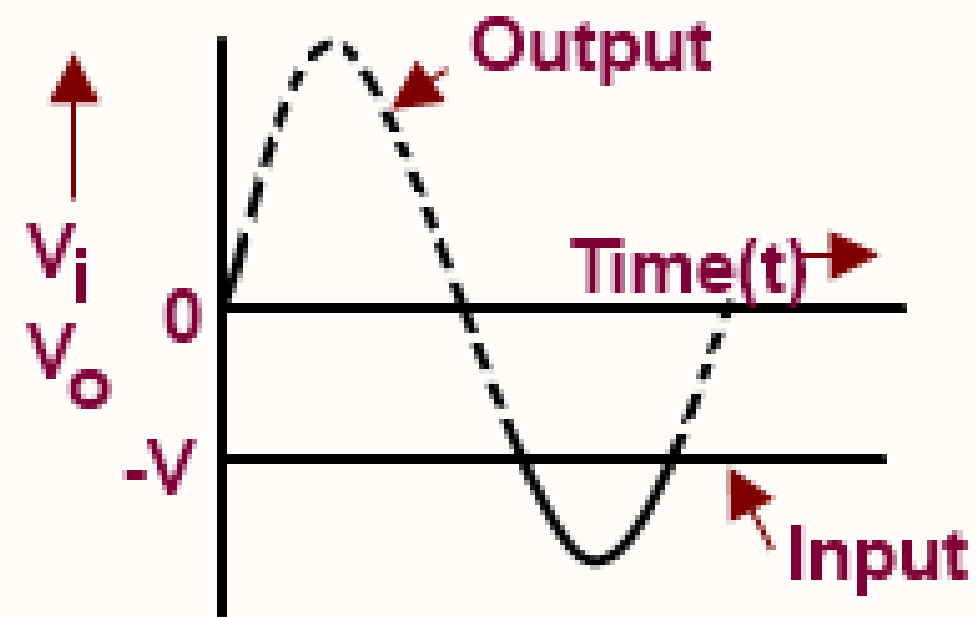
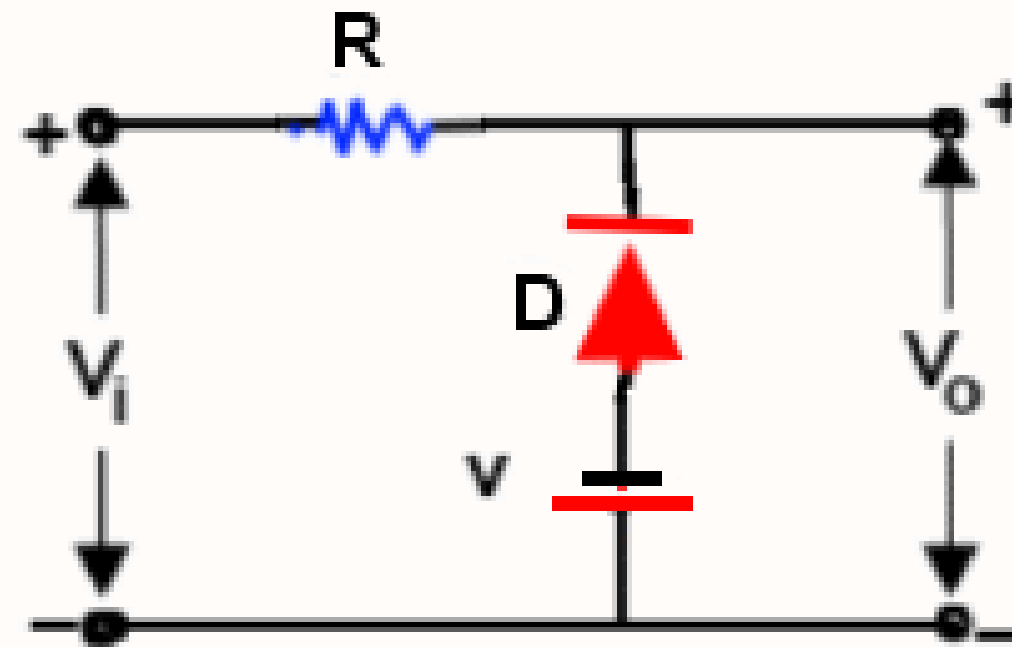
Clipper Circuits



Diode Circuit



(a) Positive bias

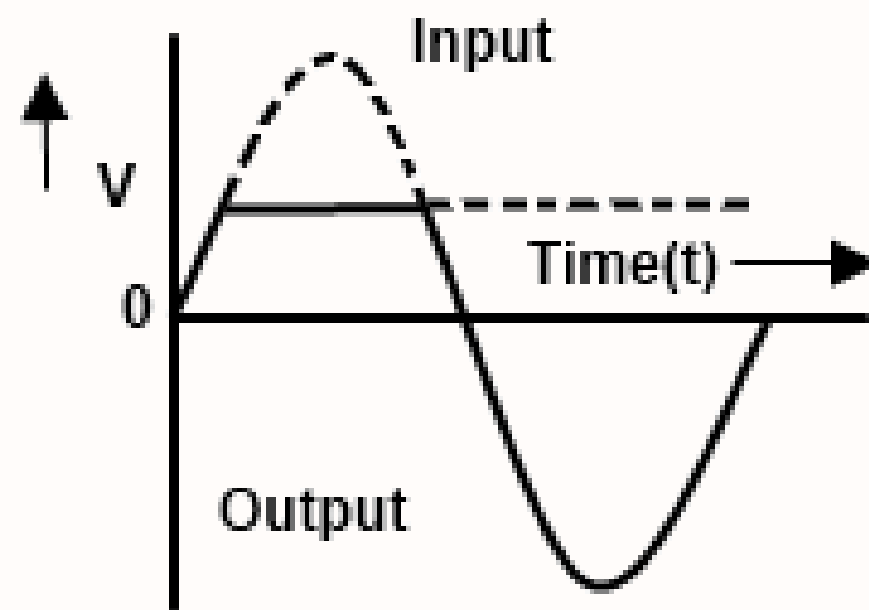
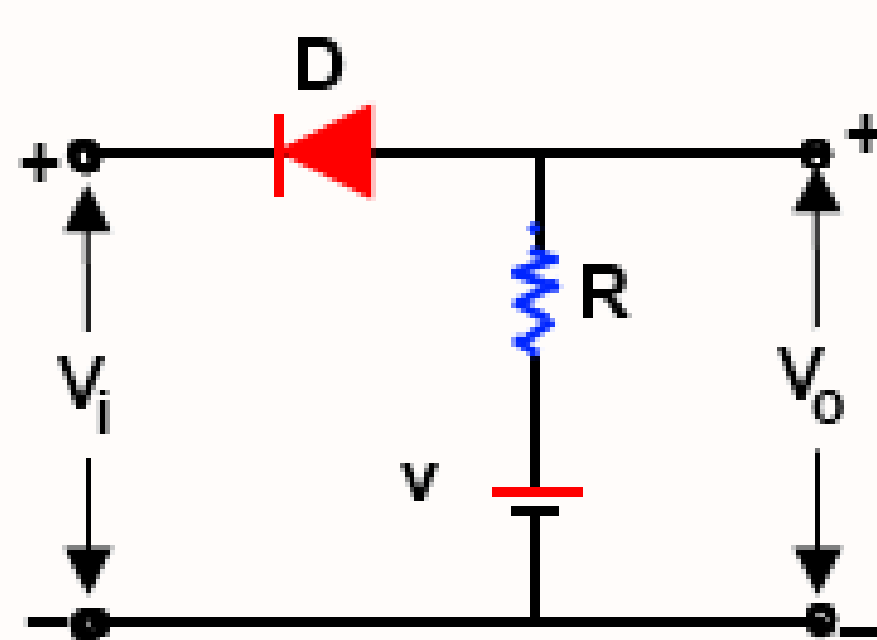


(b) negative bias

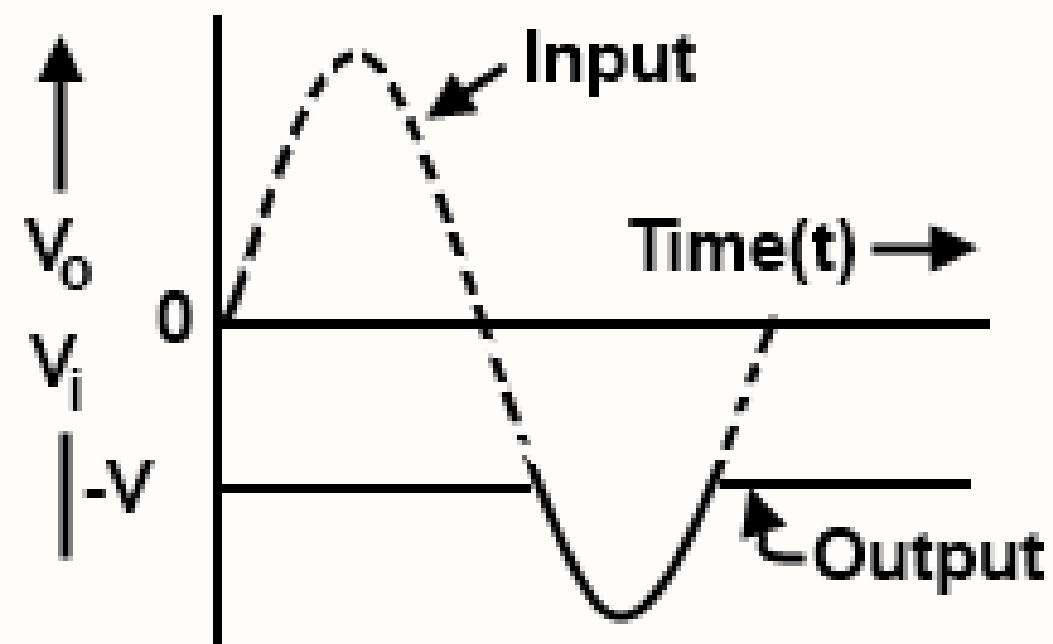
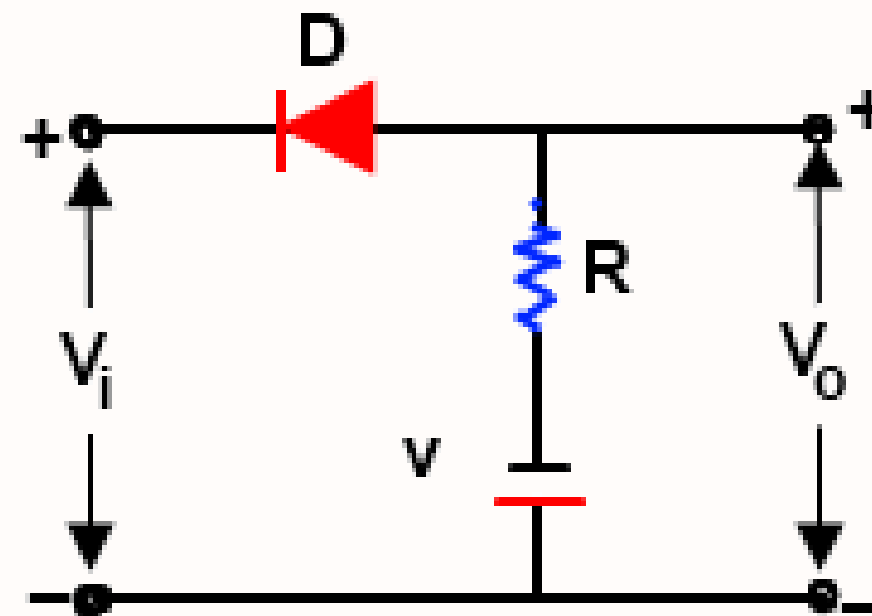
Clipper Circuits



Diode Circuit



(a) Positive Biased

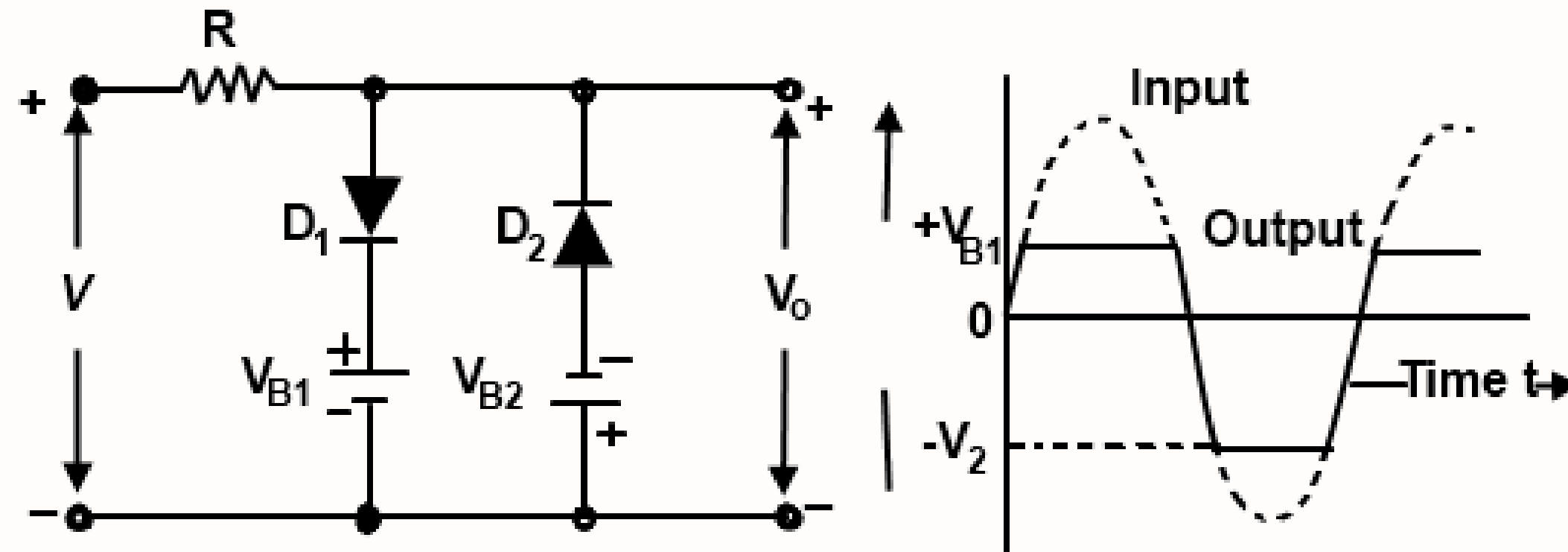


(b) Negative Biased

Clipper Circuits



Diode Circuit



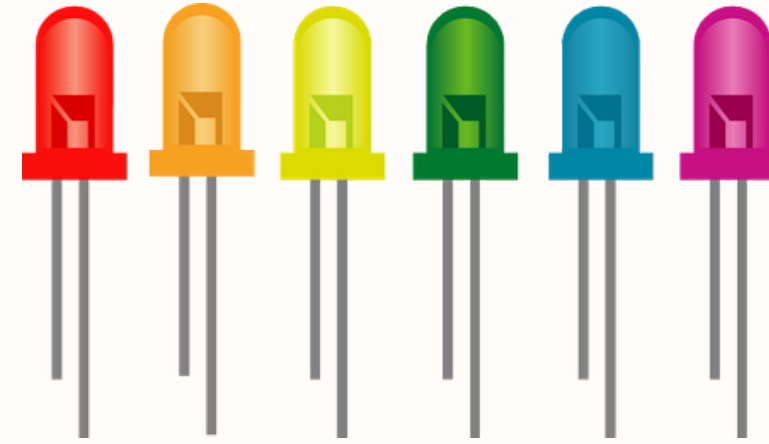
Limiter



Other types of diode

For Examples:

1. Light Emitting Diode (LED)

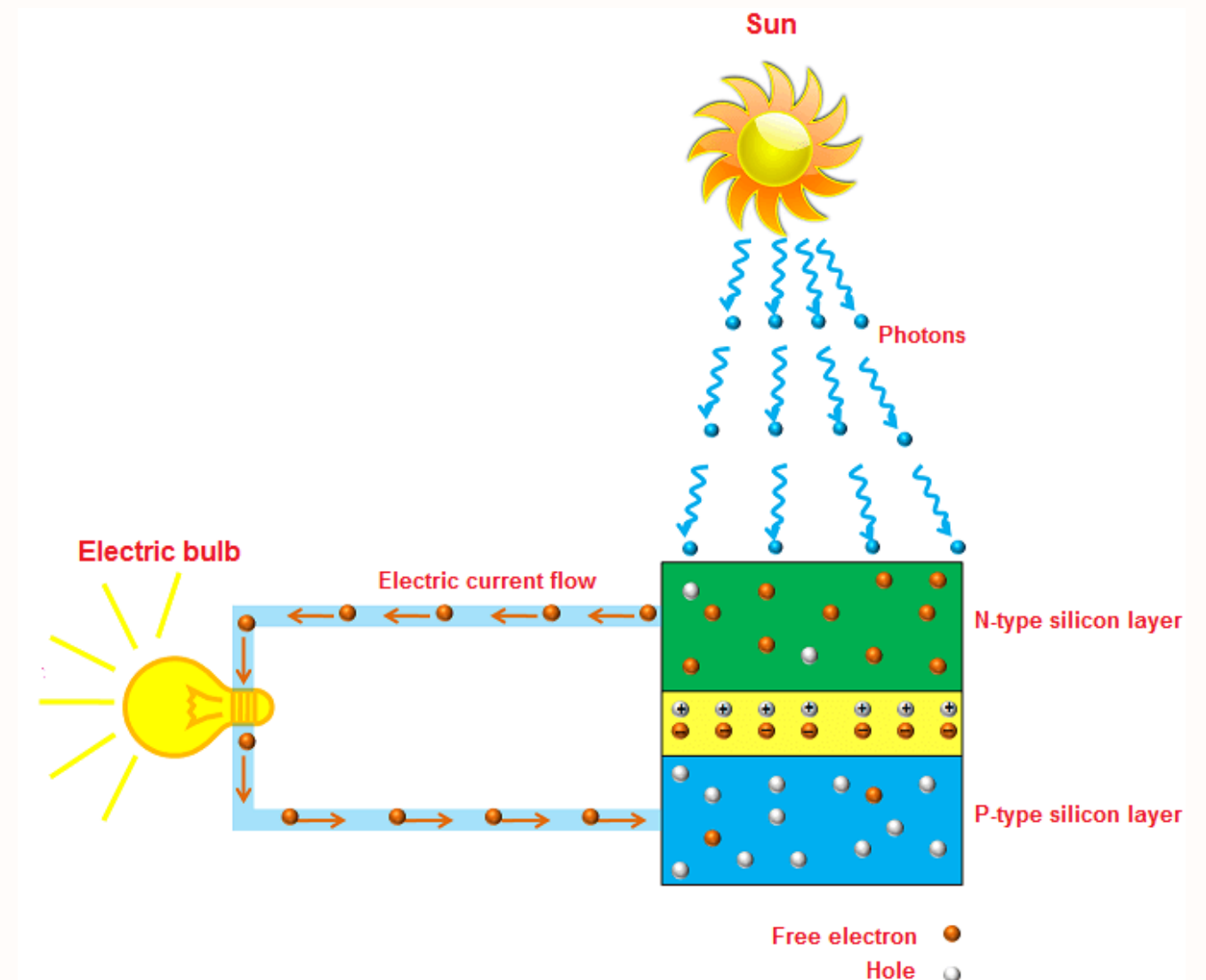


2. Photodiode



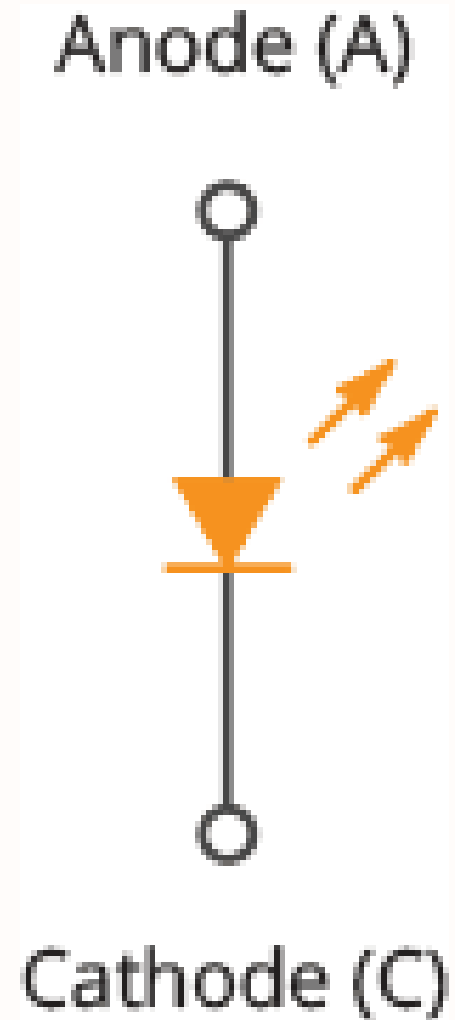
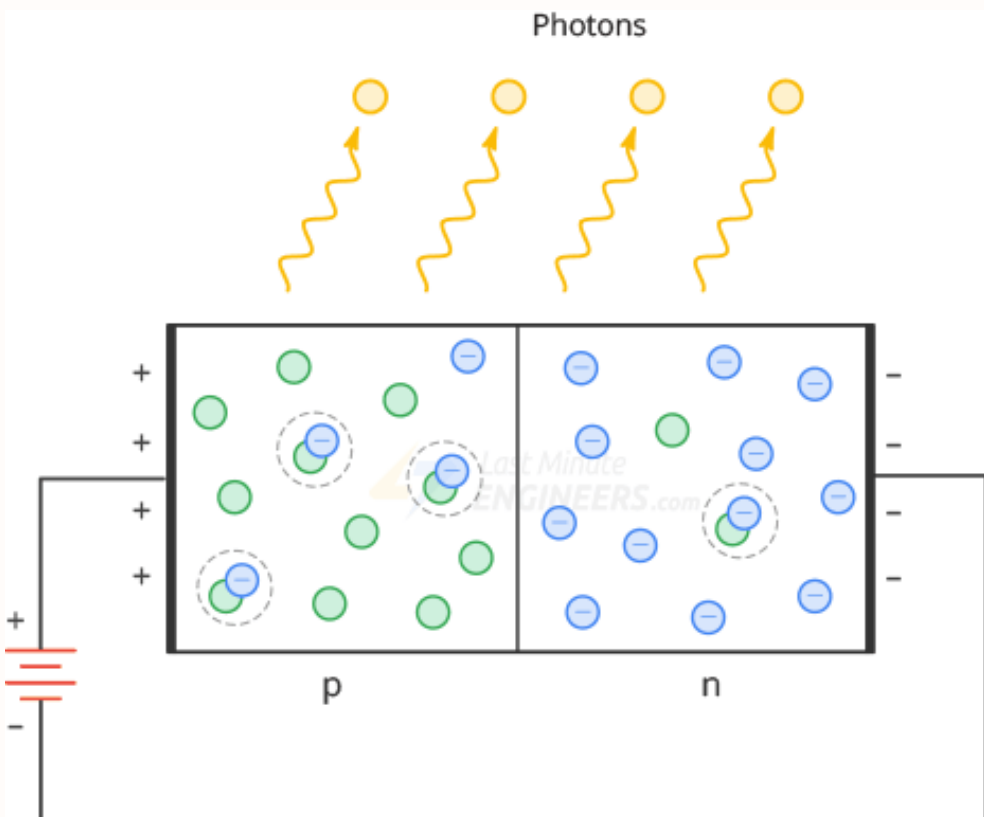
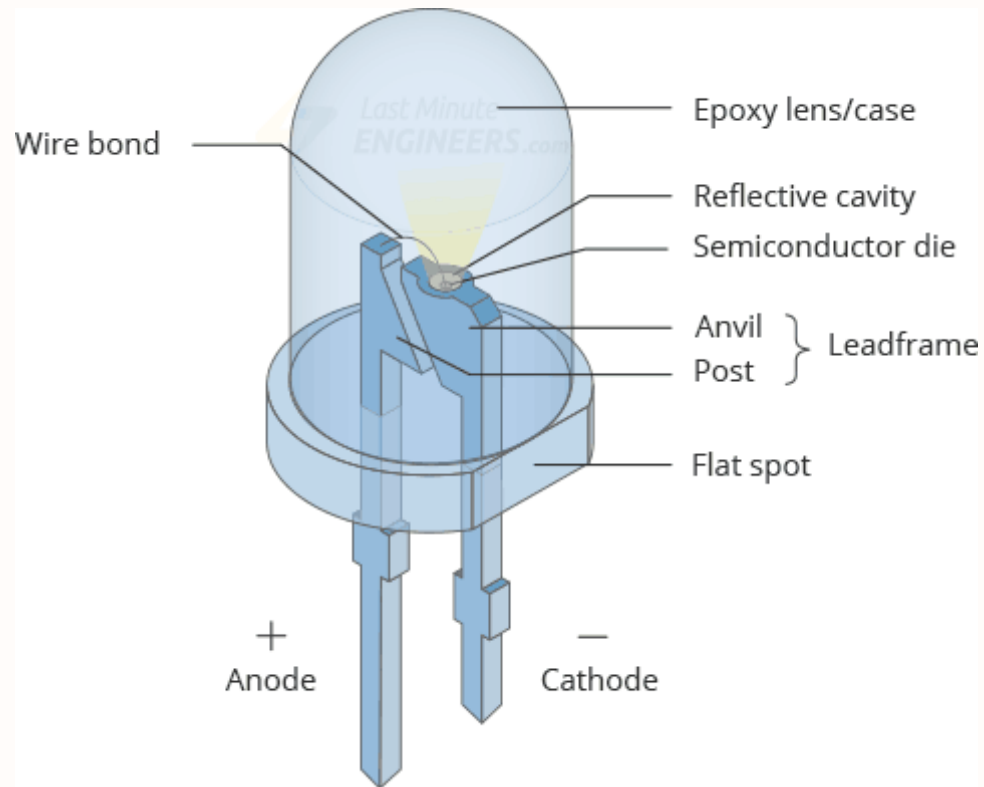
3. Zener diode

4. Solar cell





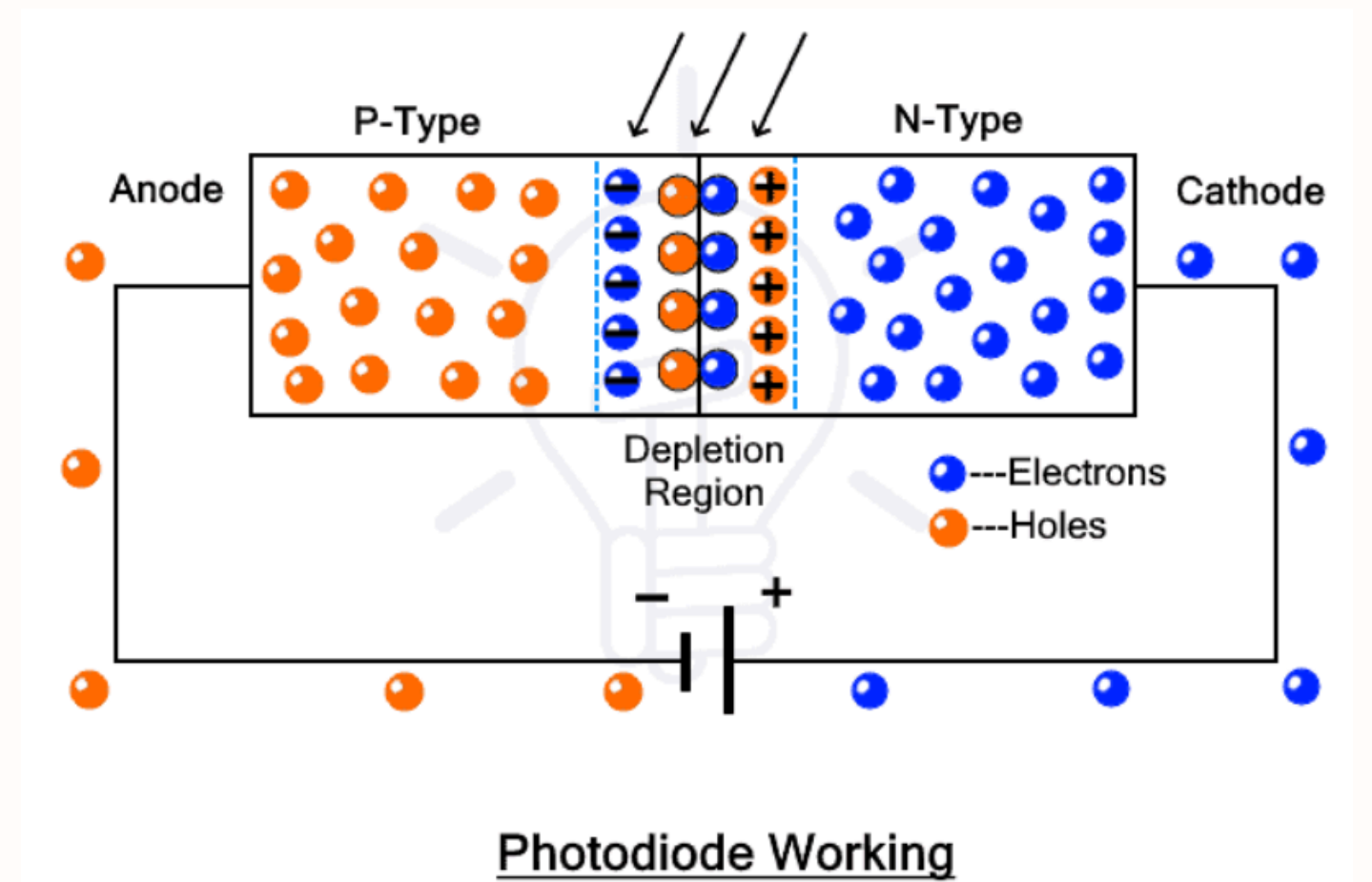
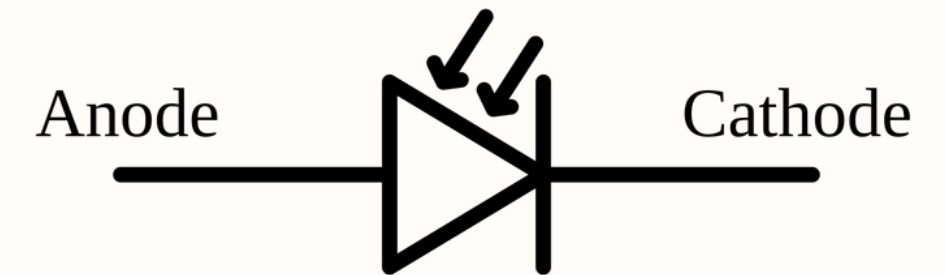
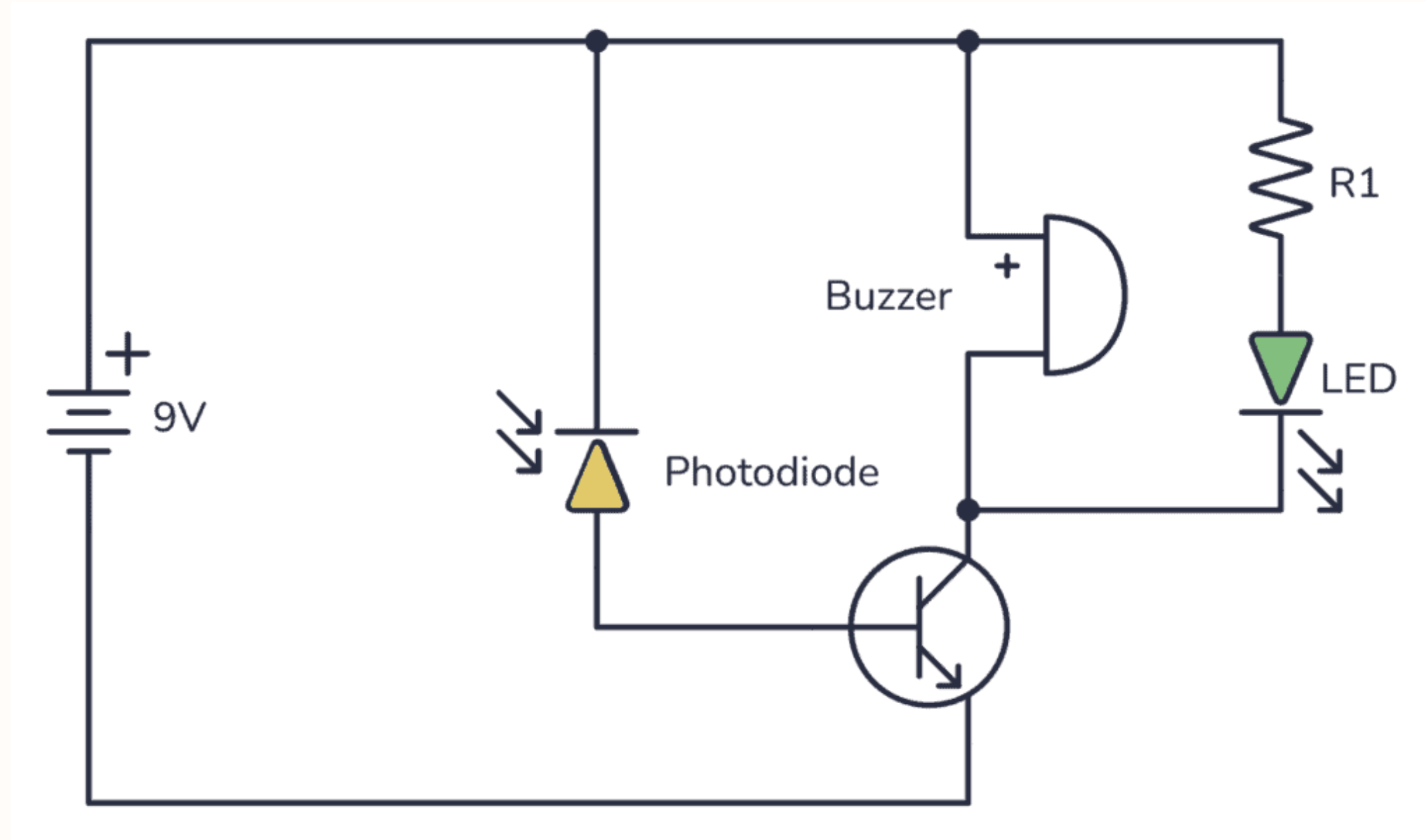
LED



Color	Wavelength (nm)	Forward Voltage (V)	Material
 Ultraviolet	<400	3.1-4.4	Aluminium nitride(ALN) Aluminium gallium nitride (AlGaN)
 Violet	400-450	2.8-4.0	Indium gallium nitride (InGaN)
 Blue	450-500	2.5-3.7	Indium gallium nitride (InGaN) Silicon carbide (SiC)
 Green	500-570	1.9-4.0	Gallium phosphide (GaP) Aluminium gallium phosphide (AlGaP)
 Yellow	570-590	2.1-2.2	Gallium arsenide phosphide (GaAsP) gallium phosphide (GaP)
 Orange	590-610	2.0-2.1	Gallium arsenide phosphide(GaAsP) gallium phosphide (GaP)
 Red	610-760	1.6-2.0	Aluminium gallium arsenide (AlGaAs) Gallium arsenide phosphide (GaAP) Gallium phosphide (GaP)
 Infrared	>760	>1.9	Gallium arsenide(GaAs) Aluminium gallium arsenide (AlGaAs)

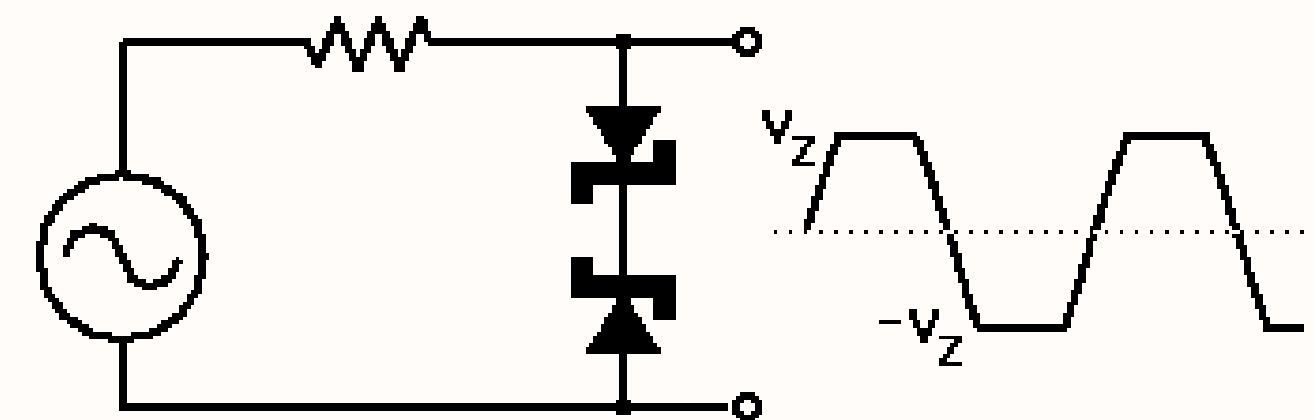
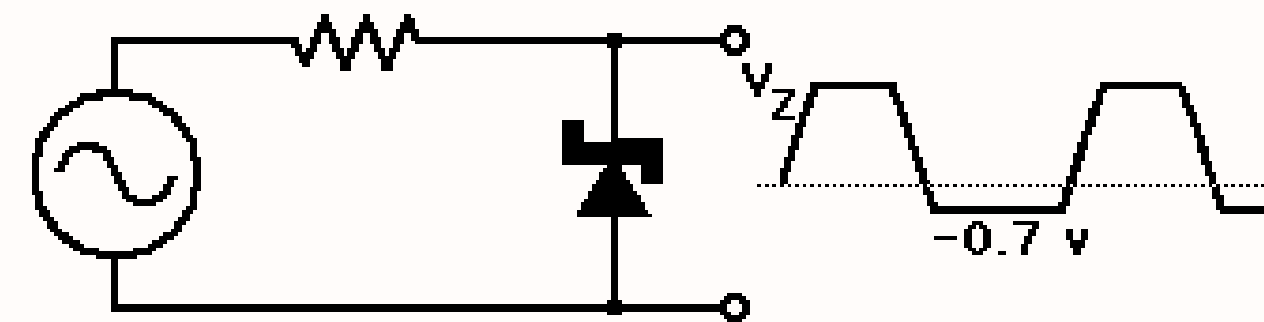
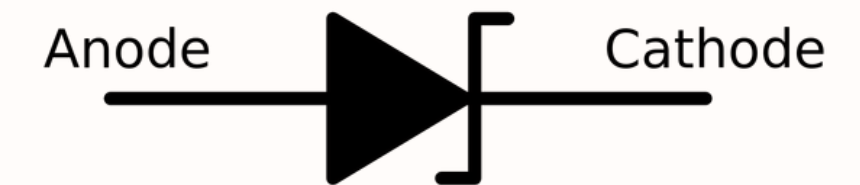
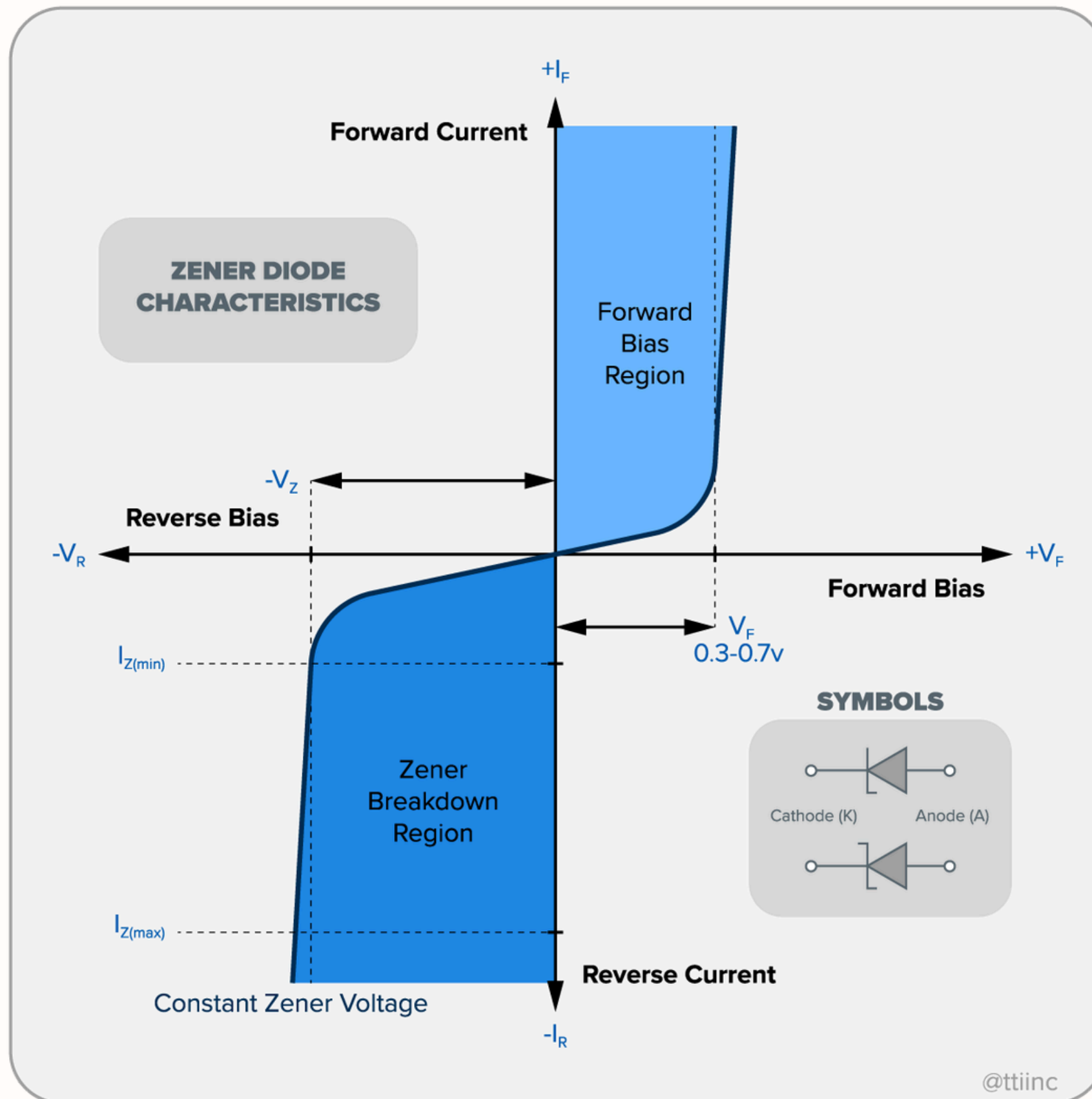


Photodiode

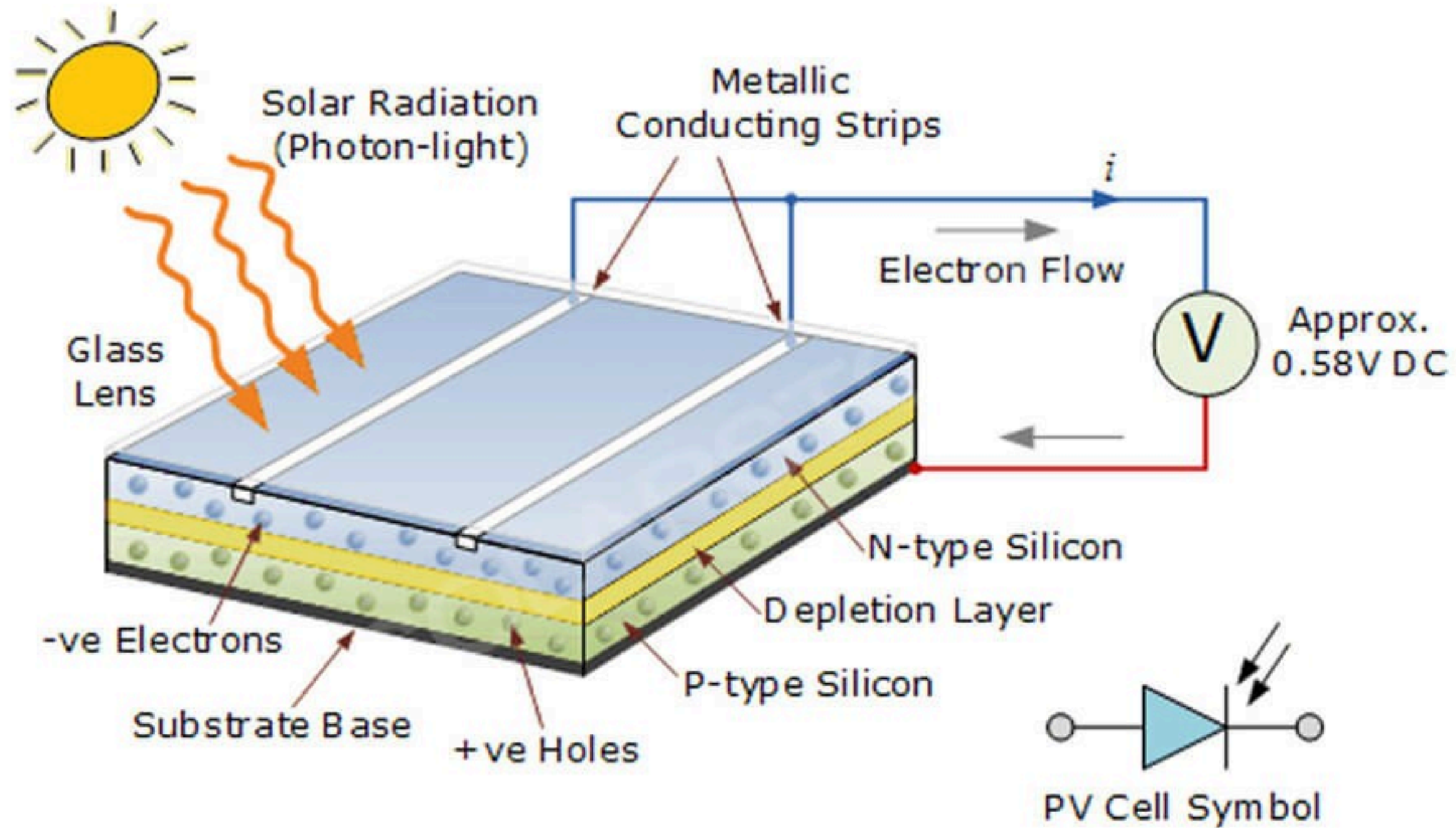




Zener diode



Solar Cell



Construction of a solar cell